ADDITION OF VECTORS

SYNOPSIS

- **Definition of Vector and Scalar:**
- **Scalar:** A physical quantity which has only magnitude is called a scalar.
  Examples: Length,volume, speed, time.
- **Vector:** A vector is a physical quantity which has both magnitude and direction . Geometrically a directed line segment is called a vector.
  Examples: Force, Velocity, acceleration.
  Note: All real numbers are scalars.
- **Notation:** Vectors are denoted by directed line segments such as $\overline{AB}\,\overline{CD}$.... or by $\overline{a},\overline{b}$ ... If $\overline{AB}$ is a vector then A is called initial point and B is called the terminal point or final point and the direction of $\overline{AB}$ is from A to B. The magnitude of $\overline{AB}$ is denoted by $|\overline{AB}|$ or AB and, is the distance between the points A and B.
- **Types of Vectors:**
- **Position Vector:** Let O be a fixed point (called the origin) and let P be any point. If $\overline{OP}=\overline{r}$ then $\overline{r}$ is called the position vector of P with respect to O.
- **Null Vector:** A vector having zero magnitude (arbitrary direction) is called the null (zero) vector. It is denoted by $\overline{0}$.
  Note (i) A zero vector can be regarded as having any direction for all mathematical calculations.
  (ii) A non-zero vector is called a proper vector
- **Unit Vector:** A vector whose magnitude is equal to one unit is called a unit vector.
- **Localised vector** :A vector is a localised vector, if the vector is specified by giving either initial point or terminal point (or) if a vector is specified by fixing atleast one of its ends is called a localised vector.
- **Free Vector**: When a vector is specified by not fixing initial point or terminal point or both, is called a free vector, i.e., a free vector does not have specific initial point or terminal point or both.
  Note: (i) A vector $\overline{a}$ means we are free to choose initial or terminal point anywhere. Once initial point is fixed at A then terminal point is uniquely fixed at B such that $\overline{AB}=\overline{a}$
  (ii) A free vector is subjected to parallel displacement without changing the magnitude and direction.
  (iii) In general vectors are considered to be free vectors unless they are localised.
- Let $\overline{a}$ be a nonzero vector then
  (i) Unit vector in the direction of $\overline{a}=\dfrac{\overline{a}}{|\overline{a}|}$
  (ii) Unit vector in the direction opposite to that of $\overline{a}$ is $-\dfrac{\overline{a}}{|\overline{a}|}$
  (iii) Unit vectors parallel to $\overline{a}=\pm\dfrac{\overline{a}}{|\overline{a}|}$.
  (iv) The vectors having magnitude m units and parallel to $\overline{a}=\pm\dfrac{m\overline{a}}{|\overline{a}|}$.

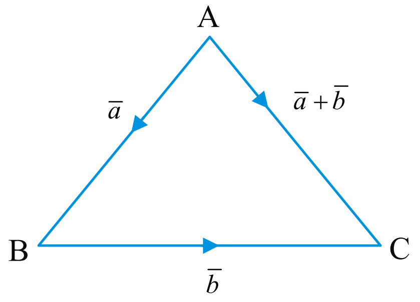
WE-1: ***$\overline{a}=\overline{i}+2\overline{j}+2\overline{k}$ and $\overline{b}=3\overline{i}+6\overline{j}+2\overline{k}$, then vector in the direction of $\overline{a}$ and having magnitude as $|\overline{b}|$ is***

Sol: The required vector is

$$\frac{|\overline{b}|}{|\overline{a}|}\overline{a}=\frac{7}{3}\left(\overline{i}+2\overline{j}+2\overline{k}\right)$$

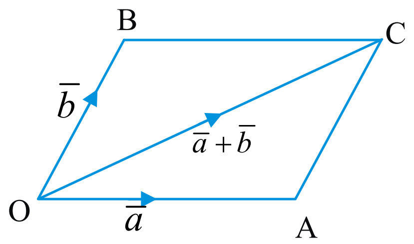
- **Equal Vectors:** Two vectors $\overline{a}$ and $\overline{b}$ are equal if they have the same magnitude $\left(i.e\,|\overline{a}|=|\overline{b}|\right)$ and they are in the same direction.
- Let $\overline{a}$ and $\overline{b}$ be the position vectors of the points A and B respectively. Then $\overline{AB}=$ (position vector of B) - (position vector of A) i.e., $\overline{AB}=\overline{b}-\overline{a}$

- **Co-Initial Vectors:** Vectors having the same initial point are called coinitial vectors. The vectors $\overline{OA}, \overline{OB}, \overline{OC}...$ are coinitial vectors.
- **Co-Terminal Vectors:** The vectors having the same terminal point are called the co-terminal vectors. The vectors $\overline{AO}, \overline{BO}, \overline{CO}...$ are co-terminal vectors.
- **Addition of Vectors:**
- **Triangle law of Vector Addition:**



If $\overline{AB}=\overline{a}$ and $\overline{BC}=\overline{b}$ are two non-zero vectors are represented by two sides of a triangle ABC then the resultant (sum) vector is given by the closing side $\left(\overline{AC}\right)$ of the triangle in opposite direction. i.e., $\overline{AC}=\overline{AB}+\overline{BC}=\overline{a}+\overline{b}$

- **Parallelogram Law of Vector Addition:**



If $\overline{a}$ and $\overline{b}$ are two adjacent sides of the parallelogram, then their sum (resultant) $\overline{a}+\overline{b}$ represents the diagonal of the parallelogram through the common points. It is known as parallelogram law of vector addition.

i.e., $\overline{OC}=\overline{OA}+\overline{OB}=\overline{a}+\overline{b}$

- **Properties of Addition of Vectors:**

i) Addition of vectors is commutative

i.e., $\overline{a}+\overline{b}=\overline{b}+\overline{a}$

ii) Addition of vectors is associative

i.e., $\left(\overline{a}+\overline{b}\right)+\overline{c}=\overline{a}+\left(\overline{b}+\overline{c}\right)$

iii) There exists a vector $\overline{0}$ such that $\overline{a}+\overline{0}=\overline{0}+\overline{a}=\overline{a}$. Then $\overline{0}$ is called the additive identity vector.

iv) To each vector $\overline{a}$ there exists a vector $-\overline{a}$ such that $\overline{a}+(-\overline{a})=(-\overline{a})+\overline{a}=\overline{0}$. Then $-\overline{a}$ is called the additive inverse of $\overline{a}$.

- **Scalar Multiplication of Vectors:** Let $\overline{a}$ be a nonzero vector and let m be a scalar. Then m$\overline{a}$ is a scalar multiplication of $\overline{a}$ by m.

Note: i) The direction of m$\overline{a}$ is along $\overline{a}$ if m>o.
ii) The direction of m$\overline{a}$ is opposite to that of $\overline{a}$ if m<o.

- **Properties of Scalar Multiplication of Vectors:** If $\overline{a},\overline{b}$ are vectors & m, n are scalars, then the magnitude (length) of m$\overline{a}$ is $|m|$ times that of $\overline{a}$.

Note: (i) $m(n\overline{a})=n(m\overline{a})=(mn)\overline{a}$

(ii) $(m+n)\overline{a}=m\overline{a}+n\overline{a}$

(iii) $m(\overline{a}+\overline{b})=m\overline{a}+m\overline{b}$

- If $\overline{a}$ and $\overline{b}$ are any two vectors, then

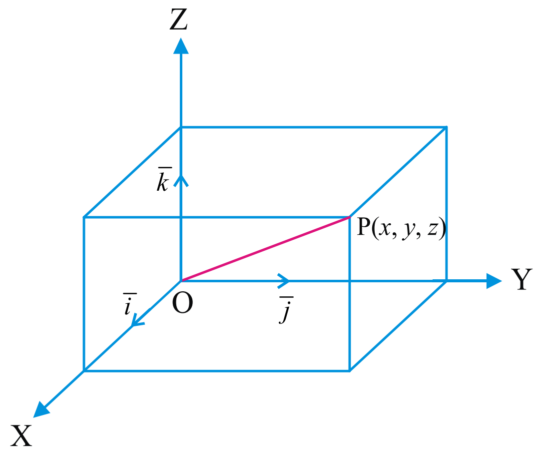
(i) $\left|\overline{a}+\overline{b}\right|\le\left|\overline{a}\right|+\left|\overline{b}\right|$

(ii) $\left|\overline{a}-\overline{b}\right|\le\left|\overline{a}\right|+\left|\overline{b}\right|$

(iii) $\left|\overline{a}-\overline{b}\right|\ge\left|\left|\overline{a}\right|-\left|\overline{b}\right|\right|$

(iv) If $\overline{a}$ and $\overline{b}$ are like vectors, then $\left|\overline{a}+\overline{b}\right|=\left|\overline{a}\right|+\left|\overline{b}\right|$

- **Components of a space Vector:**



- Let $\bar{i}, \bar{j}, \bar{k}$ be unit vectors acting along the positive directions of x, y and z axes respectively, then position vector of any point P in the space is $\overline{OP} = x\bar{i} + y\bar{j} + z\bar{k}$. Here (x, y, z) are called scalar components of vector $\overline{OP}$ along respective axes and $x\bar{i}, y\bar{j}, z\bar{k}$ are called vector components of $\overline{OP}$ along respective axes & $\left|\overline{OP}\right| = \sqrt{x^2 + y^2 + z^2}$

- **Section Formula :** If the position vectors of the points A, B w.r.t. O are $\bar{a}$ and $\bar{b}$ and if the point C divides the line segment $\overline{AB}$ in the ratio $m:n$ internally $(m>0, n>0)$, then the position vector of C is $\overline{OC} = \dfrac{m\bar{b} + n\bar{a}}{m+n}$.

- If C is an external point that divides $A(\bar{a}), B(\bar{b})$ in the ratio $m:n$ externally then
$$\overline{OC} = \frac{m\bar{b} - n\bar{a}}{m-n}\ (m \neq n) \text{ and } (m,n>0)$$

- The point C (mid point) divides $A(\bar{a}), B(\bar{b})$ in the ratio $1:1$, then $\overline{OC} = \dfrac{\bar{a}+\bar{b}}{2}$.

- **Points of trisection :** Two points which divide a line segment in the ratio $1:2$ and $2:1$ are called the points of trisection.

WE-2: If $\bar{a}, \bar{b}$ are the position vectors of A,B respectively and C is a point on AB produced such that AC = 3AB, then the position vector of C is

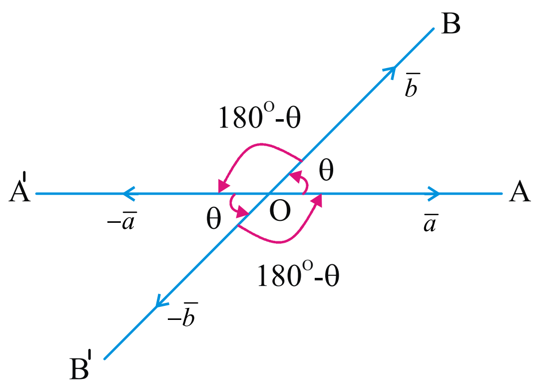
Sol: Let the position vector for C be $\bar{c}$, then B divides AC internally in the ratio 1:2, therefore
$$\bar{b} = \frac{2\bar{a} + \bar{c}}{2+1} \Rightarrow \bar{c} = 3\bar{b} - 2\bar{a}.$$

- The position vector of the centroid G of the triangle ABC with vertices $\bar{a}, \bar{b}, \bar{c}$ is $\dfrac{\bar{a}+\bar{b}+\bar{c}}{3}$ and the incentre $I = \dfrac{a\bar{a} + b\bar{b} + c\bar{c}}{a+b+c}$, where $a = BC, b = CA$ and $c = AB$.

- In ΔABC if D, E, F are the mid points of the sides BC, CA, AB respectively and G is the centroid then (i) $\overline{GA} + \overline{GB} + \overline{GC} = \bar{0}$
(ii) $\overline{AD} + \overline{BE} + \overline{CF} = \bar{0}$.
(iii) $\overline{OA} + \overline{OB} + \overline{OC} = \overline{OD} + \overline{OE} + \overline{OF} = 3\overline{OG}$

- If $\bar{a}, \bar{b}, \bar{c}$ and $\bar{d}$ are the position vectors of the vertices A, B, C and D respectively of a tetrahedron ABCD then the position vector of its centroid is $\dfrac{\bar{a}+\bar{b}+\bar{c}+\bar{d}}{4}$

- **Angle between two vectors :** If $\overline{OA} = \bar{a}, \overline{OB} = \bar{b}$ be two non-zero vectors and $\angle AOB = \theta,\ 0^0 \le \theta \le 180^0$ is defined as the angle between $\bar{a}$ and $\bar{b}$ and is written as $(\bar{a}, \bar{b})$.



Note: If $\bar{a} = \bar{0}$ or $\bar{b} = \bar{0}$, then angle between $\bar{a}$ and $\bar{b}$ is undefined.

Note: (i) $(\bar{a}, \bar{b}) = 0 \Leftrightarrow \bar{a}$ and $\bar{b}$ are like vectors.

(ii) $(\bar{a}, \bar{b}) = \dfrac{\pi}{2} \Leftrightarrow \bar{a}$ and $\bar{b}$ are orthogonal vectors.

(iii) $(\bar{a}, \bar{b}) = \pi \Leftrightarrow \bar{a}$ and $\bar{b}$ are unlike vectors.

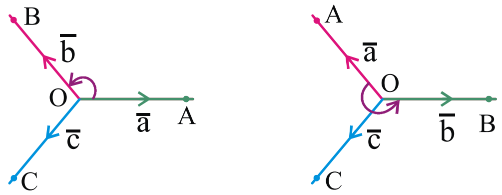
(iv) $(\bar{a}, \bar{b}) = (\bar{b}, \bar{a})$ and $(\bar{a}, \bar{b}) = (-\bar{a}, -\bar{b})$

(v) $(m\bar{a}, n\bar{b}) = (\bar{a}, \bar{b})$ (if m,n have same signs).

(vi) $(m\bar{a}, n\bar{b}) = 180^0 - (\bar{a}, \bar{b})$
(if m,n have opposite signs).

➢ **Right handed and left handed triads**

Let $\overline{OA}=\overline{a},\overline{OB}=\overline{b},\overline{OC}=\overline{c}$ be three non-coplanar vectors. Viewing from the point C, if the rotation of $\overline{OA}$ to $\overline{OB}$ does not exceed angle $180°$ in anti-clock sense, then $\overline{a},\overline{b},\overline{c}$ are set to form a **right handed system of vectors** and we say simply that $(\overline{a},\overline{b},\overline{c})$ is right handed vector triad. If $(\overline{a},\overline{b},\overline{c})$ is not a right handed system, then it is called **left handed system.**



➢ **Direction cosines and Direction ratios of a vector:** Let $\overline{i},\overline{j},\overline{k}$ be an unit vector traid in the right handed system and $\overline{r}$ is a vector. If $\alpha=(\overline{r},\overline{i}),\beta=(\overline{r},\overline{j})$ and $\gamma=(\overline{r},\overline{k})$, then $\cos\alpha,\cos\beta,\cos\gamma$ are called the direction cosines of $\overline{r}$ denoted by l,m,n respectively. The numbers proportional to direction cosines of a given vector, i,e., kl,km,kn are called the direction ratios of that vector for $k\in R-\{0\}$

➢ **Some important results**: If l,m,n are the direction cosines of a line, then

$l^2+m^2+n^2=1$.

➢ If $\overline{OP}=\overline{r}$ and P is the ordered triad (x,y,z) then $x=r\cos\alpha=lr, y=r\cos\beta=mr$ and $z=r\cos\gamma=nr$.

➢ The direction cosines of the vectors $\overline{i},\overline{j},\overline{k}$ are respectively $(1,0,0),(0,1,0),(0,0,1)$.

➢ **Linear Combination of Vectors:**

➢ **Linear Combination:** Let $\overline{a}_1,\overline{a}_2,...,\overline{a}_n$ be n vectors and let $\overline{r}$ be any vector. Then $\overline{r}=x_1\overline{a}_1+x_2\overline{a}_2+x_3\overline{a}_3+....+x_n\overline{a}_n$ is called a linear combination of the vectors $\overline{a}_1,\overline{a}_2,...,\overline{a}_n$.

➢ **Collinear Vectors:** Vectors which lie on a line or on parallel lines are called collinear vectors (whatever be their magnitudes).

Note:(i) Two vectors $\overline{a}$ and $\overline{b}$ are collinear if and only if $\overline{a}=m\overline{b}$ or $\overline{b}=n\overline{a}$ where m ,n are scalars (real numbers).

(ii) Let the vectors $\overline{a}=a_1\overline{i}+a_2\overline{j}+a_3\overline{k}$, $\overline{b}=b_1\overline{i}+b_2\overline{j}+b_3\overline{k}$ are collinear if and only if

$\frac{a_1}{b_1}=\frac{a_2}{b_2}=\frac{a_3}{b_3}$

(iii) The Vectors $\overline{a},\overline{b}$ are collinear vectors iff $(\overline{a},\overline{b})=0^0$ or $180°$

➢ Let $\overline{a}$ and $\overline{b}$ be two non collinear vectors and let $\overline{r}$ be any vector coplanar with them. Then $\overline{r}=x\overline{a}+y\overline{b}$ and the scalars x and y are unique in the sense that if $\overline{r}=x_1\overline{a}+y_1\overline{b}$ and $\overline{r}=x_2\overline{a}+y_2\overline{b}$ then $x_1=x_2$ and $y_1=y_2$. The vector equation $\overline{r}=x\overline{a}+y\overline{b}$ implies that the vectors $\overline{r},\overline{a},\overline{b}$ are coplanar.

➢ Let $\overline{a}$ and $\overline{b}$ be two noncollinear vectors. If $l_1\overline{a}+m_1\overline{b}=l_2\overline{a}+m_2\overline{b}$ then $l_1=l_2$ and $m_1=m_2$.

➢ If $\overline{a}$ and $\overline{b}$ are two noncollinear vectors and $x\overline{a}+y\overline{b}=\overline{0}$ then the scalars $x=0$ and $y=0$.

➢ **Collinearity of Three Points:** Let $A(\overline{a}),B(\overline{b}),C(\overline{c})$ be three points. A necessary and sufficient condition for these points to be collinear is that there exist three scalars x,y,z not all zero such that $x\overline{a}+y\overline{b}+z\overline{c}=\overline{0}$ and $x+y+z=0$

➢ The points $A(\overline{a}),B(\overline{b}),C(\overline{c})$ are collinear if and only if $\overline{AB}=m\overline{AC}$ or $\overline{AC}=n\overline{AB}$,where m and n are scalars.

WE-3 : The points with position vectors $60\overline{i}+3\overline{j},40\overline{i}-8\overline{j},a\overline{i}-52\overline{j}$ are collinear then a = ___.

Sol: Let P,Q and R be the collinear points with position vectors $60\overline{i}+3\overline{j},40\overline{i}-8\overline{j}$ and $a\overline{i}-52\overline{j}$ respectively. Then $\overline{PQ}=\lambda\overline{QR}$ for some scalar λ

$\Rightarrow -20\overline{i}-11\overline{j}=\lambda((a-40)\overline{i}-44\overline{j})$

$\Rightarrow \lambda(a-40)=-20,-44\lambda=-11$

$\Rightarrow \lambda=1/4$ and $\lambda(a-40)=-20$

$\Rightarrow a-40=-80\Rightarrow a=-40$

W.E-4: *Let $\bar{a}$, $\bar{b}$ and $\bar{c}$ be three non-zero vectors which are pairwise non-collinear. If $\bar{a}+3\bar{b}$ is collinear with $\bar{c}$ and $\bar{b}+2\bar{c}$ is collinear with $\bar{a}$, then $\bar{a}+3\bar{b}+6\bar{c}$ is __ [AIE-2011]*

1) $\bar{a}+\bar{c}$ 2) $\bar{a}$ 3) $\bar{c}$ 4) $\bar{0}$

Sol: As $\bar{a}+3\bar{b}$ is collinear with $\bar{c}$, $\bar{a}+3\bar{b}=\lambda\bar{c}$... (1)

As $\bar{b}+2\bar{c}$ is collinear with $\bar{a}$, $\bar{b}+2\bar{c}=\mu\bar{a}$...(2)

from (1) we get $\bar{a}+3\bar{b}+6\bar{c}=(\lambda+6)\bar{c}$...(3)

from (2) we get $\bar{a}+3\bar{b}+6\bar{c}=(1+3\mu)\bar{a}$...(4)

since $\bar{a}$ is not collinear with $\bar{c}$,

$\lambda+6=1+3\mu=0$

from (3) or (4) $\bar{a}+3\bar{b}+6\bar{c}=\bar{0}$

➤ **Coplanar Vectors:** Vectors which lie on a plane or parallel to the same plane are called coplanar vectors. The vectors $\bar{a},\bar{b},\bar{c}$ are coplanar if their exist scalar x,y,z not all zero such that $x\bar{a}+y\bar{b}+z\bar{c}=\bar{0}$.

➤ A necessary and sufficient condition for four points $A(\bar{a}), B(\bar{b}), C(\bar{c}), D(\bar{d})$ to be coplanar is that there exist four scalars x,y,z,t not all zero such that $x\bar{a}+y\bar{b}+z\bar{c}+t\bar{d}=\bar{0}$ and $x+y+z+t=0$

➤ Let $\bar{a},\bar{b},\bar{c}$ be three noncoplanar vectors then the vector equation $x\bar{a}+y\bar{b}+z\bar{c}=\bar{0}$ implies that $x=y=z=0$.

➤ Let $\bar{a},\bar{b},\bar{c}$ three non-coplanar vectors. Then any vector $\bar{r}$ can be expressed as a linear combination of $\bar{a},\bar{b},\bar{c}$ i.e.,

$\bar{r}=x\bar{a}+y\bar{b}+z\bar{c}$. Further the scalars x, y, z are unique in the sense that if

$\bar{r}=x_1\bar{a}+y_1\bar{b}+z_1\bar{c}$ and,

$\bar{r}=x_2\bar{a}+y_2\bar{b}+z_2\bar{c}$ then

$x_1=x_2, y_1=y_2$ and $z_1=z_2$.

WE-5 : *If $\bar{a},\bar{b},\bar{c}$ are three non-coplanar vectors such that $\bar{a}+\bar{b}+\bar{c}=\alpha\bar{d}$ and $\bar{b}+\bar{c}+\bar{d}=\beta\bar{a}$ then $\bar{a}+\bar{b}+\bar{c}+\bar{d}$ is ___*

Sol: We have,

$\bar{a}+\bar{b}+\bar{c}=\alpha\bar{d}$ and $\bar{b}+\bar{c}+\bar{d}=\beta\bar{a}$

Let $\bar{a}+\bar{b}+\bar{c}+\bar{d}=(\alpha+1)\bar{d}$ and

$\bar{a}+\bar{b}+\bar{c}+\bar{d}=(\beta+1)\bar{a}$

$\Rightarrow(\alpha+1)\bar{d}=(\beta+1)\bar{a}$, If $\alpha\neq-1$, then

$(\alpha+1)\bar{d}=(\beta+1)\bar{a}\Rightarrow\bar{d}=\dfrac{\beta+1}{\alpha+1}\bar{a}$

$\therefore\bar{a}+\bar{b}+\bar{c}=\alpha\bar{d}$

$\Rightarrow\bar{a}+\bar{b}+\bar{c}=\alpha\left(\dfrac{\beta+1}{\alpha+1}\right)\bar{a}$

$\Rightarrow\left(1-\dfrac{\alpha(\beta+1)}{\alpha+1}\right)\bar{a}+\bar{b}+\bar{c}=\bar{0}$

$\Rightarrow\bar{a},\bar{b},\bar{c}$ are coplanar. Which is a contradiction to the given condition

$\therefore\alpha=-1\Rightarrow\bar{a}+\bar{b}+\bar{c}+\bar{d}=\bar{0}$

➤ **Linearly Dependent (L.D) and Independent (L.I) Vectors:** A system of vectors $\bar{a}_1,\bar{a}_2,...,\bar{a}_n$ is linearly dependent if there exists a system of scalars $x_1,x_2,...,x_n$ not all zero (atleast one of the scalar is non zero) such that $x_1\bar{a}_1+x_2\bar{a}_2+...+x_n\bar{a}_n=\bar{0}$

➤ The system of vectors $\bar{a}_1,\bar{a}_2,...,\bar{a}_n$ is linearly independent if every relation of the form $x_1\bar{a}_1+x_2\bar{a}_2+...+x_n\bar{a}_n=\bar{0}$ implies that $x_1=x_2=..........=x_n=0$

➤ **Note:** i) Any two non collinear vectors are L.I.
ii) Any two collinear vectors are L.D.
iii) Any three non co-planar vectors are L.I.
iv) Any three co-planar vectors are L.D.
v) Any four vectors in 3D-space are L.D.
vi) Any super set of L.D. system of vectors is also L.D.
vii) Any Sub set of L.I. system of vectors is also L.I.

WE-6: *If the vectors $\bar{a}$ and $\bar{b}$ are linearly independent satisfying $\left(\sqrt{3}\tan\theta+1\right)\bar{a}+\left(\sqrt{3}\sec\theta-2\right)\bar{b}=\bar{0}$, then the most general values of θ are*

Sol: $\sqrt{3}\tan\theta+1=0$ and $\sqrt{3}\sec\theta-2=0$

$\Rightarrow\theta=\dfrac{11\pi}{6}\Rightarrow\theta=2n\pi+\dfrac{11\pi}{6}, n\in Z$

➢ To verify that three given vectors $\bar{a}=a_1\bar{i}+a_2\bar{j}+a_3\bar{k}$, $\bar{b}=b_1\bar{i}+b_2\bar{j}+b_3\bar{k}$ and $\bar{c}=c_1\bar{i}+c_2\bar{j}+c_3\bar{k}$ are linearly independent or linearly dependent,

find $\Delta=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

i) If $\Delta \neq 0$, then they are linearly independent.
ii) If $\Delta=0$, then they are linearly dependent.

➢ Definition of Orthonormal Vectors:

If $\bar{i},\bar{j},\bar{k}$ are three vectors such that $|\bar{i}|=|\bar{j}|=|\bar{k}|=1$ and $(\bar{i},\bar{j})=(\bar{j},\bar{k})=(\bar{k},\bar{i})=90^0$ then $\bar{i},\bar{j},\bar{k}$ are called Orthonormal Vectors

➢ Vector Equation of Lines:

(i) Vector equation to a line passing through the point A($\bar{a}$) and parallel to the vector $\bar{b}$ is

$\bar{r}=\bar{a}+t\bar{b}\ (t\in R)$

(ii) Vector equation to a line passing through two points A($\bar{a}$) and B($\bar{b}$) is $\bar{r}=(1-t)\bar{a}+t\bar{b}$ $(t\in R)$

(iii) Vector equation to a line passing through the origin (position vector $\bar{0}$) and parallel to the vector $\bar{a}$ is $\bar{r}=t\bar{a}$. $(t\in R)$

➢ (i) The cartesian equation to a line passing through the point $A=(x_1,y_1,z_1)$ and parallel to the vector (a,b,c) is $\dfrac{x-x_1}{a}=\dfrac{y-y_1}{b}=\dfrac{z-z_1}{c}$.

(ii) The cartesian equation to a line passing through two points $A=(x_1,y_1,z_1)$ and B$=(x_2,y_2,z_2)$ is $\dfrac{x-x_1}{x_1-x_2}=\dfrac{y-y_1}{y_1-y_2}=\dfrac{z-z_1}{z_1-z_2}$ or

$\dfrac{x-x_2}{x_1-x_2}=\dfrac{y-y_2}{y_1-y_2}=\dfrac{z-z_2}{z_1-z_2}$

(iii) The cartesian equation of a line passing through the origin (0,0,0) and the point (x_1,y_1,z_1) is $\dfrac{x}{x_1}=\dfrac{y}{y_1}=\dfrac{z}{z_1}$.

➢ (i) The vector equation to the X-axis is $\bar{r}=\lambda\bar{i}$, λ is a scalar.

(ii) The cartesian equation to the X-axis is $\dfrac{x}{1}=\dfrac{y}{0}=\dfrac{z}{0}$.

➢ Vector Equation of a Plane:

(i) Vector equation to a plane passing through the point $A(\bar{a})$ and parallel to the vectors $\bar{b}$ and $\bar{c}$ is $\bar{r}=\bar{a}+t\bar{b}+s\bar{c}\ (t,s\in R)$

(ii) Vector equation to a plane passing through the points $A(\bar{a})$ and $B(\bar{b})$ and parallel to $\bar{c}$ is

$\bar{r}=(1-t)\bar{a}+t\bar{b}+s\bar{c}\ \ (t,s\in R)$

(iii) Vector equation to a plane passing through the points $A(\bar{a})$ and $B(\bar{b})$ and $C(\bar{c})$ is

$\bar{r}=(1-t-s)\bar{a}+t\bar{b}+s\bar{c}\ \ (t,s\in R)$

(iv)Vector equation to a plane passing through the origin (**0**) and parallel to the vectors $\bar{b}$ and $\bar{c}$ is

$\bar{r}=t\bar{b}+s\bar{c}$.(where t, s are parameters)

➢ **Standard Results:** If AD is the internal bisector of the angle A, then D divides BC in the ratio AB : AC. then

$$\overline{AD}=\frac{|\overline{AC}|\,\overline{AB}+|\overline{AB}|\,\overline{AC}}{|\overline{AB}|+|\overline{AC}|}$$

WE-7 : The vertices of a triangle are $A(1,-1,-3),B(2,1,-2)$ and $C(-5,2,-6)$. The length of the bisector of its interior angle at the vertex A is____

Sol: The bisector divides BC in the ratio AB:AC i.e $\sqrt{6}:3\sqrt{6}$ or $1:3$ at point D.Therefore the position vector of D is $\dfrac{1}{4}\bar{i}+\dfrac{5}{4}\bar{j}-3\bar{k}$ and

$\overline{AD}=\dfrac{3}{4}\bar{i}+\dfrac{9}{4}\bar{j}+0\bar{k}$.

Hence, $|\overline{AD}|=\sqrt{\dfrac{9}{16}+\dfrac{81}{16}}=\dfrac{3}{4}\sqrt{10}$

➢ Let $\overline{OA}=\bar{a}$ and $\overline{OB}=\bar{b}$. Then a vector along

a) The internal bisector of the angle $\angle AOB$ is

$\lambda\left(\frac{\bar{a}}{|\bar{a}|}+\frac{\bar{b}}{|\bar{b}|}\right)$, λ is a scalar.

b) The external bisector of the angle $\angle AOB$ is

$\lambda\left(\frac{\bar{a}}{|\bar{a}|}-\frac{\bar{b}}{|\bar{b}|}\right)$, λ is a scalar.

➢ (i) Vector equation of the internal bisector of an angle between two vectors $\bar{b}$ and $\bar{c}$ with vertex at $\bar{a}$ is $\bar{r}=\bar{a}+t\left(\frac{\bar{b}}{|\bar{b}|}+\frac{\bar{c}}{|\bar{c}|}\right)$, where t is a scalar.

(ii) Vector equation of the external angular bisector is $\bar{r}=\bar{a}+t\left(\frac{\bar{b}}{|\bar{b}|}-\frac{\bar{c}}{|\bar{c}|}\right)$ where t is a scalar.

WE- 8 : If the vector $-\bar{i}+\bar{j}-\bar{k}$ bisects the angle between the vector $\bar{c}$ and the vector $3\bar{i}+4\bar{j}$,then the unit vector in the direction of $\bar{c}$ is ____

Sol : Let $x\bar{i}+y\bar{j}+z\bar{k}$ be the unit vector along $\bar{c}$. Since $-\bar{i}+\bar{j}-\bar{k}$ bisects the angle between $\bar{c}$ and $3\bar{i}+\bar{j}$. Therefore,

$$\lambda\left(-\bar{i}+\bar{j}-\bar{k}\right)=\left(x\bar{i}+y\bar{j}+z\bar{k}\right)+\frac{3\bar{i}+4\bar{j}}{5}$$

$$\Rightarrow x+\frac{3}{5}=-\lambda, y+\frac{4}{5}=\lambda \text{ and } z=-\lambda$$

Now, $x^2+y^2+z^2=1$

($\because x\bar{i}+y\bar{j}+z\bar{k}$ is a unit vector)

$$\Rightarrow\left(-\lambda-\frac{3}{5}\right)^2+\left(\lambda-\frac{4}{5}\right)^2+\lambda^2=1$$

$$\Rightarrow \lambda=0 \text{ or } \lambda=\frac{2}{15}.$$

But $\lambda\neq 0$. Because $\lambda=0$ implies that the given vectors are parallel.

$$\therefore \lambda=\frac{2}{15}\Rightarrow x=-\frac{11}{15}, y=\frac{-10}{15} \text{ and } z=\frac{-2}{15}$$

Hence,

$$x\bar{i}+y\bar{j}+z\bar{k}=-\frac{1}{15}\left(11\bar{i}+10\bar{j}+2\bar{k}\right).$$

WE-9 : The vector $\bar{c}$,directed along the internal bisector of the angle between the vectors $\bar{a}=7\bar{i}-4\bar{j}-4\bar{k}$ and $\bar{b}=-2\bar{i}-\bar{j}+2\bar{k}$ with $|\bar{c}|=5\sqrt{6}$, is ____

Sol : The required vector $\bar{c}$ is given by $\bar{c}=\lambda\left(\frac{\bar{a}}{|\bar{a}|}+\frac{\bar{b}}{|\bar{b}|}\right)$

$$=\lambda\left\{\frac{1}{9}\left(7\bar{i}-4\bar{j}-4\bar{k}\right)+\frac{1}{3}\left(-2\bar{i}-\bar{j}+2\bar{k}\right)\right\}$$

$$\text{or, } \bar{c}=\frac{\lambda}{9}\left(\bar{i}-7\bar{j}+2\bar{k}\right)$$

$$\Rightarrow|\bar{c}|=\pm\frac{\lambda}{9}\sqrt{1+49+4}=\pm\frac{\lambda}{9}\sqrt{54}$$

But $|\bar{c}|=5\sqrt{6}$ (given).

$$\therefore \pm\frac{\lambda}{9}\sqrt{54}=5\sqrt{6}\Rightarrow\lambda=\pm 15$$

Hence,

$$\bar{c}=\pm\frac{15}{9}\left(\bar{i}-7\bar{j}+2\bar{k}\right)=\pm\frac{5}{3}\left(\bar{i}-7\bar{j}+2\bar{k}\right)$$

➢ i) If $A(a_1,b_1,c_1)$, $B(a_1,b_1,c_2)$ and $C(a_1,b_2,c_1)$ are the vertices of a ΔABC then,

(i) Circumcentre is $=\left(a_1,\frac{b_1+b_2}{2},\frac{c_1+c_2}{2}\right)$

(ii) Orthocentre is $A\left(a_1,b_1,c_1\right)$

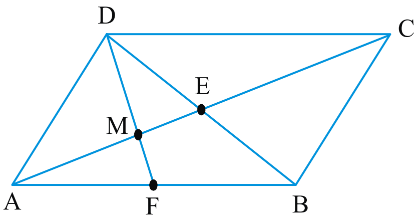
➢ In ΔABC if 'S' is the circumcentre and 'O' is the orthocentre then

(i) $\overline{SA}+\overline{SB}+\overline{SC}=\overline{SO}$

(ii) $\overline{OA}+\overline{OB}+\overline{OC}=2\,\overline{OS}$

(iii) $\overline{AO}+\overline{OB}+\overline{OC}=2\overline{AS}$

➢ **Parallelogram:** Let ABCD be a Parallelogram



i) Diagonal $\overline{AC} = \overline{AB} + \overline{BC}$ and diagonal $\overline{BD} = \overline{BC} + \overline{CD} = -\overline{AB} + \overline{BC}$

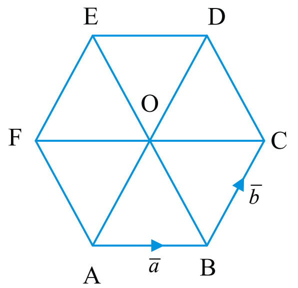
ii) AC and BD are diagonals of a parallelogram then, a) $\overline{AB} = \dfrac{\overline{AC} - \overline{BD}}{2}$ b) $\overline{BC} = \dfrac{\overline{AC} + \overline{BD}}{2}$

iii) If E is the point of intersection of diagonals then $\overline{OA} + \overline{OB} + \overline{OC} + \overline{OD} = 4\,\overline{OE}$ (diagonals bisect each other)

iv) If F is midpoint of AB and DF intersects AC at M then (i) AM : MC = 1 : 2 and (ii) DM : MF = 2:1.

➢ **Regular Hexagon:**

Consider regular hexagon ABCDEF.



a) If $\overline{AB} = \bar{a}$ and $\overline{BC} = \bar{b}$ then

i) $\overline{AC} = \overline{AB} + \overline{BC} = \bar{a} + \bar{b}$

ii) $\overline{AD} = 2\overline{BC} = 2\bar{b}$ iii) $\overline{CD} = \bar{b} - \bar{a}$

iv) $\overline{DE} = -\overline{AB} = -\bar{a}$ v) $\overline{EF} = -\overline{BC} = -\bar{b}$

vi) $\overline{FA} = -\overline{CD} = \bar{a} - \bar{b}$

b) If 'O' is the centre of the hexagon then

$\overline{AB}+\overline{AC}+\overline{AD}+\overline{AE}+\overline{AF}=3\overline{AD}=6\overline{AO}$

➢ Let A,B be two points Such that $\bar{a} = \overline{OA}$, $\bar{b} = \overline{OB}$ than the point C where $\overline{OC} = p\bar{a} + q\bar{b}$ lies

(i) inside ΔOAB, if p>0, q>0 and p+q<1

(ii) outside ΔOAB but inside $\angle AOB$ if p>0, q>0 and p+q>1

(iii) outside ΔOAB but inside $\angle OAB$ if p<0, q>0 and p+q<1

(iv) outside ΔOAB but inside $\angle OBA$ if p>0, q<0 and p+q <1

(v) outside ΔOAB if p<0, q>0 and p+q>1

Rotation of a vector about an axis:

➢ Let $\bar{a} = (a_1, a_2, a_3)$. If the system is rotated about

i) x-axis through an angle α, then the new components of $\bar{a}$ are

$(a_1, a_2\cos\alpha + a_3\sin\alpha, -a_2\sin\alpha + a_3\cos\alpha)$

ii) y-axis through an angle α, then the new components of $\bar{a}$ are

$(-a_3\sin\alpha + a_1\cos\alpha, a_2, a_3\cos\alpha + a_1\sin\alpha)$

iii) z-axis through an angle α, then the new components of $\bar{a}$ are

$(a_1\cos\alpha + a_2\sin\alpha, -a_1\sin\alpha + a_2\cos\alpha, a_3)$

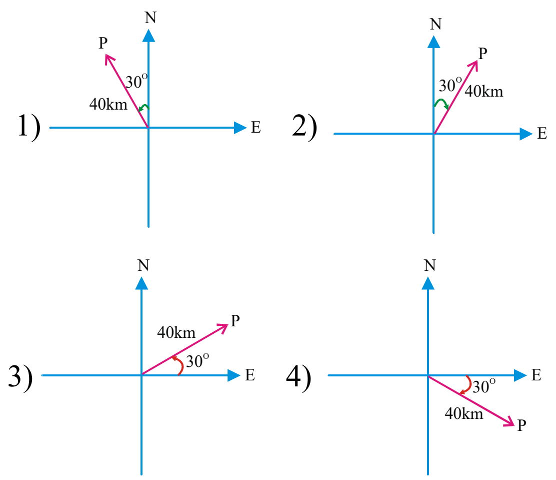
➢ **Ceva's theorem:** If the lines joining any point P to the vertices A,B,C of a triangle meet the opposite sides in D,E,F respectively,then

$$\frac{BD}{DC}\bullet\frac{CE}{EA}\bullet\frac{AF}{FB} = 1$$

➢ **Menelaus' theorem:** If a transversal cuts the sides BC,CA,AB of a triangle in D,E,F respectively,then $\dfrac{BD}{DC}\bullet\dfrac{CE}{EA}\bullet\dfrac{AF}{FB} = -1$

C.U.Q

1. Which of the following diagram graphically represents a displacement of 40 km, 30^0 east of north



2. Which of the following is true

1) $\bar{a}$ and $-\bar{a}$ are collinear
2) Two collinear vectors are always equal in magnitude
3) Two vectors having same magnitude are collinear
4) Two collinear vectors having the same magnitude are equal

3. If $\bar{a}$ and $\bar{b}$ are two non-collinear vectors and $x\bar{a}+y\bar{b}=\bar{0}$, Then

1) $x=0$, but y is not neccessarily zero
2) $y=0$, but x is not necessarily zero
3) $x=0, y=0$ 4) $x\neq 0, y\neq 0$

4. If $\bar{a}$ and $\bar{b}$ are two collinear vectors then which of the following is incorrect ?

1) $x\bar{a}+y\bar{b}=\bar{0}$ and $x+y=0$
2) $\bar{b}=\lambda\bar{a}$, λ is a scalar
3) both the vectors have the same direction, but different magnitudes
4) the respective components of $\bar{a}$ and $\bar{b}$ are proportional

5. If the points $A(\bar{a}),B(\bar{b}),C(\bar{c})$ satisfy the relation $3\bar{a}-8\bar{b}+5\bar{c}=0$ then the points are

1) vertices of an equilateral triangle
2) collinear
3) vertices of a right angled triangle
4) vertices of an isosceles triangle

6. If the points A, B, C, D have the position vectors $\bar{a},\bar{b},\bar{c}$ and $\bar{d}$ respectively such that $(\lambda+\mu+\gamma)\bar{a}=\lambda\bar{b}+\mu\bar{c}+\gamma\bar{d}$ then which of the following is incorrect.

1) Lines AC and BD are parallel
2) Points A, B, C, D are collinear
3) Points A, B, C, D are non collinear
4) Lines AB&CD are parallel

7. Let α,β,γ be distinct real numbers. The points with position vectors $\alpha\bar{i}+\beta\bar{j}+\gamma\bar{k}$, $\beta\bar{i}+\gamma\bar{j}+\alpha\bar{k}$, $\gamma\bar{i}+\alpha\bar{j}+\beta\bar{k}$

1) are collinear
2) form an equilateral triangle
3) form a scalene triangle
4) form a rightangled triangle

8. If $\overline{AD},\overline{BE},\overline{CF}$ are medians of an equilateral triangle ABC, then $\overline{AD}+\overline{BE}+\overline{CF}=$

1) $\overline{AB}+\overline{BC}+\overline{CA}$ 2) a zero vector
3) both 1 and 2 4) $2\overline{AF}+3\overline{BF}$

9. In a trapezium, the straight line joining the mid points of the diagonals is half the

1) difference of parallel sides
2) sum of parallel sides
3) difference of non parallel sides
4) sum of non parallel sides

10. If ABCD is a Rhombus whose diagonals cut at the origin *O* then $\overline{OA}+\overline{OB}+\overline{OC}+\overline{OD}=$

1) $\overline{AB}$ 2) $\overline{O}$ 3) $\overline{AD}$ 4) $\overline{BC}$

11. If the parallelogram ABCD with $\overline{AC}$ as the diagonal is completed, then the position vector of the point D is

1) $\frac{1}{2}(\bar{C}+\bar{A}-\bar{B})$ 2) $\frac{1}{3}(\bar{A}+\bar{B}-\bar{C})$
3) $\bar{A}+\bar{C}-\bar{B}$ 4) $\bar{A}+\bar{B}-\bar{C}$

12. Three non-zero, non-parallel coplanar vectors are always

1) linearly dependent 2) linearly independent
3) either of (1) or (2) 4) cannot be determined

13. If OABC is a parallelogram with $\overline{OB}=\bar{a}$, $\overline{AB}=\bar{b}$ then $\overline{OA}=$ **(EAM- 2000)**

1) $\bar{a}+\bar{b}$ 2) $\bar{a}-\bar{b}$
3) $\frac{1}{2}(\bar{b}-\bar{a})$ 4) $\frac{1}{2}(\bar{a}-\bar{b})$

14. If $A(\bar{a}),B(\bar{b})$ and $C(\bar{c})$ be the vertices of a triangle whose circumcentre is the origin then orthocentre is given by

1) $\bar{a}+\bar{b}+\bar{c}$ 2) $\frac{\bar{a}+\bar{b}+\bar{c}}{3}$
3) $\frac{\bar{a}+\bar{b}+\bar{c}}{2}$ 4) $\frac{\bar{a}+\bar{b}+\bar{c}}{4}$

15. Let $\bar{a},\bar{b},\bar{c}$ be three unit vectors such that $3\bar{a}+4\bar{b}+5\bar{c}=\bar{0}$ Then

1) $\bar{a},\bar{b},\bar{c}$ are collinear
2) $\bar{a},\bar{b},\bar{c}$ are pair wise orthogonal
3) $\bar{a},\bar{b},\bar{c}$ are linearly independent
4) $\bar{a},\bar{b},\bar{c}$ are linearly dependent

16. If $\bar{a}$ and $\bar{b}$ are two non-collinear vectors, then the points $l_1\bar{a}+m_1\bar{b}$, $l_2\bar{a}+m_2\bar{b}$ and $l_3\bar{a}+m_3\bar{b}$ are collinear if

1) $\Sigma l_1(m_2-m_3)=0$ 2) $\Sigma l_1(m_1-m_3)=0$

3) $\Sigma l_1(m_1+m_3)=0$ 4) $\Sigma l_1(m_2+m_3)=0$

17. If θ is an angle given by

$\cos\theta=\dfrac{\cos^2\alpha+\cos^2\beta+\cos^2\gamma}{\sin^2\alpha+\sin^2\beta+\sin^2\gamma}$ where α,β,γ are the angles made by a line with the positive directions of the axes of reference then the measure of θ is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{3}$

18. The direction cosines of two lines are (l_1,m_1,n_1) and (l_2,m_2,n_2). Then the value of $(l_1l_2+m_1m_2+n_1n_2)^2+\sum(m_1n_2-m_2n_1)^2=$

1) 1 2) 0 3) 4 4) 2

19. If I is the centre of a circle inscribed in a triangle ABC, then $|\overline{BC}|\overline{IA}+|\overline{CA}|\overline{IB}+|\overline{AB}|\overline{IC}$ is

1) $\bar{0}$ 2) $\overline{IA}+\overline{IB}+\overline{IC}$

3) $\dfrac{\overline{IA}+\overline{IB}+\overline{IC}}{3}$ 4) $\dfrac{\overline{IA}+\overline{IB}+\overline{IC}}{2}$

20. If $\bar{a},\bar{b}$ are non-collinear vectors, then

1) $\bar{a}+\bar{b}$ represents the bisector of the angle between $\bar{a}$ and $\bar{b}$

2) If $|\bar{a}|=|\bar{b}|$, then $\bar{a}+\bar{b}$ bisects the angles between $\bar{a}$ and $\bar{b}$

3) $\bar{a},\bar{b},\bar{a}+\bar{b}$ from an equilateral triangle

4) $\bar{a}-\bar{b},\bar{a}+\bar{b}$ are other two sides of the parallelogram for which $\bar{a},\bar{b}$ are two adjacent sides.

21. If point O is the centre of a circle circumscribed about a triangle ABC. Then $\overline{OA}\sin 2A+\overline{OB}\sin 2B+\overline{OC}\sin 2C=$

1) $(\overline{OA}+\overline{OB}+\overline{OC})\sin 2A$

2) $(\overline{OA}+\overline{OB}+\overline{OC})\cos 2A$

3) $\bar{0}$ 4) $(\overline{OA}+\overline{OB}+\overline{OC})\tan 2A$

22. The vector equation of the plane passing through $\bar{a},\bar{b},\bar{c}$ is $\bar{r}=\alpha\bar{a}+\beta\bar{b}+\gamma\bar{c}$ provided

1) $\alpha+\beta+\gamma=0$ 2) $\alpha+\beta+\gamma=1$

3) $\alpha+\beta=\gamma$ 4) $\alpha^2+\beta^2+\gamma^2=1$

C.U.Q-KEY

01) 2	02) 1	03) 3	04) 3	05) 2	06) 3
07) 2	08) 3	09) 1	10) 2	11) 3	12) 1
13) 2	14) 1	15) 4	16) 1	17) 4	18) 1
19) 1	20) 2	21) 3	22) 2		

LEVEL - I (C.W)

POSITION VECTOR,UNIT VECTOR AND EQUAL VECTORS:

1. The vector $(\cos\alpha\cos\beta)\,\bar{i}+(\cos\alpha\sin\beta)\bar{j}+\sin\alpha\,\bar{k}$ is

1) Null vector 2) unit vector

3) parallel to $(\bar{i}+\bar{j}+\bar{k})$

4) a vector parallel to $(2\bar{i}+\bar{j}-\bar{k})$

2. If $\overline{AB}=2\bar{i}-3\bar{j}+\bar{k}$, $\overline{CB}=\bar{i}+\bar{j}+\bar{k}$, $\overline{CD}=4\bar{i}-7\bar{j}$ then $\overline{AD}=$

1) $5\bar{i}+11\bar{j}-\bar{k}$ 2) $5\bar{i}-11\bar{j}$

3) $5\bar{i}+11\bar{j}$ 4) $-5\bar{i}+11\bar{j}$

3. The position vectors of the points A,B,C are $\bar{i}+2\bar{j}-\bar{k}$, $\bar{i}+\bar{j}+\bar{k}$, $2\bar{i}+3\bar{j}+2\bar{k}$ respectively. If A is chosen as the origin then the position vectors of B and C are

1) $\bar{i}+2\bar{k},\bar{i}+\bar{j}+3\bar{k}$ 2) $\bar{j}+2\bar{k},\bar{i}+\bar{j}+3\bar{k}$

3) $-\bar{j}+2\bar{k},\bar{i}-\bar{j}+3\bar{k}$

4) $-\bar{j}+2\bar{k},\bar{i}+\bar{j}+3\bar{k}$

4. The unit vector in the opposite direction of the vector $\bar{a}=-6\bar{i}+3\bar{j}-2\bar{k}$ is

1) $\frac{1}{7}(-6\bar{i}+3\bar{j}-2\bar{k})$ 2) $\frac{1}{7}(6\bar{i}-3\bar{j}-2\bar{k})$

3) $\frac{1}{7}(6\bar{i}-3\bar{j}+2\bar{k})$ 4) $\frac{1}{7}(-6\bar{i}+3\bar{j}+2\bar{k})$

5. If $\bar{a}=(2,1,-1),\bar{b}=(1,-1,0),\ \bar{c}=(5,-1,1)$ then the unit vector parallel to $\bar{a}+\bar{b}-\bar{c}$, but in the opposite direction is

1) $-\frac{1}{3}(2\bar{i}-\bar{j}+2\bar{k})$ 2) $\frac{1}{3}(2\bar{i}-\bar{j}+2\bar{k})$

3) $\frac{1}{3}(2\bar{i}+\bar{j}-2\bar{k})$ 4) $-\frac{1}{3}(2\bar{i}-\bar{j}-2\bar{k})$

COLLINEAR, LIKE AND UNLIKE VECTORS:

6. Let $\bar{a}$, $\bar{b}$ be two noncollinear vectors .If $\overline{OA}$=(x+4y)$\bar{a}$+(2x+y+1)$\bar{b}$, $\overline{OB}$=(y-2x+2)$\bar{a}$+ (2x-3y-1)$\bar{b}$ and $3\overline{OA}=2\overline{OB}$, then (x,y) =
1) (1,2) 2) (1,-2) 3) (2,-1) 4) (-2,-1)

7. If the vectors $\bar{a}=-2\bar{i}+3\bar{j}+y\bar{k}$ and $\bar{b}=x\bar{i}-6\bar{j}+2\bar{k}$ are collinear, then the value of x + y is ....
1) 4 2) 5 3) -3 4) 3

8. If the position vectors of the points P,Q,R and S are respectively $2\bar{i}+4\bar{k}$, $5\bar{i}+3\sqrt{3}\bar{j}+4\bar{k}$, $-2\sqrt{3}\bar{j}+\bar{k}$ and $2\bar{i}+\bar{k}$, then $\dfrac{|\overline{PQ}|}{|\overline{RS}|}$ is
1) $\frac{2}{3}$ 2) $\frac{3}{2}$ 3) $\frac{1}{3}$ 4) $\frac{3}{4}$

9. If the vectors $\bar{a}=2\bar{i}+3\bar{j}+6\bar{k}$ and $\bar{b}$ are collinear and $|\bar{b}|=21$, then $\bar{b}$ equals to
1) $\pm2\left(2\bar{i}+3\bar{j}+6\bar{k}\right)$ 2) $\pm3\left(2\bar{i}+3\bar{j}+6\bar{k}\right)$
3) $\bar{i}+\bar{j}+\bar{k}$ 4) $\pm21\left(2\bar{i}+3\bar{j}+6\bar{k}\right)$

10. If the vectors $\bar{a}=(x,-2,5)$ and $\bar{b}=(1,y,-3)$ are collinear then
1) $x=\frac{5}{3}, y=\frac{6}{5}$ 2) $x=\frac{5}{3}, y=\frac{-6}{5}$
3) $x=\frac{-5}{3}, y=\frac{6}{5}$ 4) $x=\frac{-5}{3}, y=\frac{-6}{5}$

11. If $\bar{p}=\bar{i}+a\bar{j}+\bar{k}$ and $\bar{q}=\bar{i}+\bar{j}+\bar{k}$, then $|\bar{p}+\bar{q}|=|\bar{p}|+|\bar{q}|$ is true for
1) $a=-1$ 2) $a=1$
3) all real values of 'a' 4) for no real values of 'a'

COPLANAR AND NON COPLANAR VECTORS:

12. If $3\bar{i}+3\bar{j}+\sqrt{3}\bar{k}$, $\bar{i}+\bar{k}$, $\sqrt{3}\bar{i}+\sqrt{3}\bar{j}+\lambda\bar{k}$ are coplanar then λ is ...
1) 1 2) 2 3) 3 4) 4

13. If the points $(\alpha-x,\alpha,\alpha),(\alpha,\alpha-y,\alpha)$, $(\alpha,\alpha,\alpha-z)$ and $(\alpha-1,\alpha-1,\alpha-1)$ are coplanar, $\alpha\in R$ then
1) $xy+yz+zx=1$ 2) $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$
3) $\frac{1}{1-x}+\frac{1}{1-y}+\frac{1}{1-z}=1$ 4) x y z = 1

14. Let $\bar{a}=\bar{i}+\bar{j}+\bar{k},\bar{b}=\bar{i}-\bar{j}+2\bar{k}$ and $\bar{c}=x\bar{i}+(x-2)\bar{j}+\bar{k}$. If the vector $\bar{c}$ lies in the plane of $\bar{a}$ and $\bar{b}$ then x equals [AIE-2007]
1) 0 2) 1 3) –4 4) –2

TRIANGLE AND PARALLELOGRAM LAW, VECTOR ADDITION

15. If the position vectors of the vertices of a triangle are $2\bar{i}-\bar{j}+\bar{k},\bar{i}-3\bar{j}-5\bar{k}$ and $3\bar{i}-4\bar{j}-4\bar{k}$ then the triangle is
1) Equilateral triangle 2) Isosceles triangle
3) Right angled isosceles triangle
4) Right angled triangle

16. If the vectors $4\bar{i}-7\bar{j}-2\bar{k},\bar{i}+5\bar{j}-3\bar{k}$ and $3\bar{i}-\lambda\bar{j}+\bar{k}$ form a triangle then $\lambda=$
1) 6 2) -6 3) 12 4) -1

17. Let ABC be a triangle and let D,E be the midpoints of the sides AB,AC respectively,then $\overline{BE}+\overline{DC}=$
1) $\overline{BC}$ 2) $\frac{1}{2}\overline{BC}$ 3) $\frac{3}{2}\overline{BC}$ 4) $\frac{3}{4}\overline{BC}$

18. Orthocentre of an equilateral triangle ABC is the origin O. If $\overline{OA}=\bar{a},\overline{OB}=\bar{b},\overline{OC}=\bar{c}$ then $\overline{AB}+2\overline{BC}+3\overline{CA}=$
1) $3\bar{c}$ 2) $3\bar{a}$ 3) $\bar{0}$ 4) $3\bar{b}$

19. ABC is a triangle and P is any point on BC. If $\overline{PQ}$ is the resultant of the vectors $\overline{AP}$, $\overline{PB}$ and $\overline{PC}$ then ACQB is
1) rectangle 2) square
3) rhombus 4) parallelogram

20. The adjacent sides of a parallelogram are $2\bar{i}+4\bar{j}-5\bar{k}$ and $\bar{i}+2\bar{j}+3\bar{k}$ then the unit vector parallel to a diagonal is

1) $\frac{-\bar{i}+2\bar{j}+8\bar{k}}{\sqrt{69}}$ 2) $\frac{3\bar{i}+6\bar{j}+2\bar{k}}{7}$

3) $\frac{-\bar{i}+2\bar{j}-8\bar{k}}{\sqrt{69}}$ 4) $\frac{-\bar{i}-2\bar{j}+8\bar{k}}{\sqrt{69}}$

SCALAR MULTIPLICATION AND THEIR PROPERTIES:

21. The vector $\bar{i}+x\bar{j}+3\bar{k}$ is rotated through an angle θ and doubled in magnitude, then it becomes $4\bar{i}+(4x-2)\bar{j}+2\bar{k}$. The value of x is

1) $\left\{-\frac{2}{3},2\right\}$ 2) $\left\{\frac{1}{3},2\right\}$ 3) $\left\{\frac{2}{3},0\right\}$ 4) $\{2,7\}$

ANGLE BETWEEN TWO VECTORS

22. If the position vectors of P and Q are $\bar{i}+2\bar{j}-7\bar{k}$ and $5\bar{i}-3\bar{j}+4\bar{k}$ respectively then the cosine of the angle between $\overline{PQ}$ and z-axis is

1) $\frac{4}{\sqrt{162}}$ 2) $\frac{11}{\sqrt{162}}$ 3) $\frac{5}{\sqrt{162}}$ 4) $\frac{-5}{\sqrt{162}}$

23. If a vector $\bar{a}$ of magnitude 50 is collinear with vector $\bar{b}=6\bar{i}-8\bar{j}-\frac{15}{2}\bar{k}$ and makes an acute angle with positive z-axis then

1) $\bar{a}=4\bar{b}$ 2) $\bar{a}=-4\bar{b}$ 3) $\bar{b}=4\bar{a}$ 4) $\bar{a}=-2\bar{b}$

SECTION FORMULA,MID POINT

24. If C is the mid point of AB and P is any point out side AB then (AIE-2005)

1) $\overline{PA}+\overline{PB}+2\overline{PC}=\bar{0}$ 2) $\overline{PA}+\overline{PB}+\overline{PC}=\bar{0}$

3) $\overline{PA}+\overline{PB}=2\overline{PC}$ 4) $\overline{PA}+\overline{PB}=\overline{PC}$

25. If $\bar{a}$ and $\bar{b}$ are position vectors of A and B respectively, then the position vector of a point C in $\overline{AB}$ produced such that $\overline{AC}=2015\,\overline{AB}$ is

1) $2014\bar{a}-2015\bar{b}$ 2) $2014\bar{b}+2015\bar{a}$

3) $2015\bar{b}+2014\bar{a}$ 4) $2015\bar{b}-2014\bar{a}$

26. If $3\bar{a}+4\bar{b}-7\bar{c}=\bar{0}$ then the ratio in which $C(\bar{c})$ divides the join of $A(\bar{a})$ and $B(\bar{b})$ is

1) 1 : 2 2) 2 : 3 3) 3 : 2 4) 4 : 3

27. The ratio in which $\bar{i}+2\bar{j}+3\bar{k}$ divides the join of $-2\bar{i}+3\bar{j}+5\bar{k}$ and $7\bar{i}-\bar{k}$ is

1) -3 : 2 2) 1 : 2 3) 2 : 3 4) -4 : 3

28. A point $C=\frac{5\bar{a}+4\bar{b}-5\bar{c}}{3}$ divides the line joining $A=\bar{a}-2\bar{b}+3\bar{c}$ and B in the ratio 2:1, then $\overline{AB}$ is

1) $\bar{a}+3\bar{b}-4\bar{c}$ 2) $2\bar{a}-3\bar{b}+4\bar{c}$

3) $\bar{a}+5\bar{b}-7\bar{c}$ 4) $2\bar{a}+3\bar{b}-4\bar{c}$

29. The position vectors of the points A,B,C are respectively $\bar{a},\bar{b},\bar{c}$. If P divides $\overline{AB}$ in the ratio 3:4 and Q divides $\overline{BC}$ in the ratio 2:1 both externally then $\overline{PQ}=$

1) $\bar{b}+\bar{c}-2\bar{a}$ 2) $2(\bar{b}+\bar{c}-2\bar{a})$

3) $4\bar{a}-\bar{b}-\bar{c}$ 4) $\frac{-2\bar{a}-\bar{b}-\bar{c}}{2}$

LINEAR COMBINATION,LINEARLY INDEPENDENT AND DEPENDENT VECTORS, DC'S & DR'S

30. If a straight line makes an angle $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ with each of the positive x, y and z-axis, a vector parallel to that line is

1) $\bar{i}$ 2) $\bar{i}+\bar{j}$ 3) $\bar{j}+\bar{k}$ 4) $\bar{i}+\bar{j}+\bar{k}$

31. If $\bar{e}=l\bar{i}+m\bar{j}+n\bar{k}$ is a unit vector ,the maximum value of lm + mn + nl is

1) $-\frac{1}{2}$ 2) 0 3) 1 4) $\frac{3}{2}$

32. Given $\bar{a}=\bar{i}+2\bar{j}+3\bar{k}, \bar{b}=2\bar{i}+3\bar{j}+\bar{k}$, $\bar{c}=8\bar{i}+13\bar{j}+9\bar{k}$, the linear relation among them if possible is

1) $2\bar{a}+3\bar{b}+\bar{c}=\bar{0}$ 2) $2\bar{a}-3\bar{b}+\bar{c}=\bar{0}$

3) $2\bar{a}+3\bar{b}-\bar{c}=\bar{0}$ 4) $\bar{a}+\bar{b}+\bar{c}=\bar{0}$

33. If the vectors $\bar{a}+1343\bar{b}+\bar{c}, -\bar{a}+\bar{b}+\bar{c}$ and $\bar{a}+\mu\bar{b}+2\bar{c}$ are linearly dependent then $\mu=$

1) 2014 2) 2015 3) 2016 4) 0

LH & RH SYSTEMS-GEOMETRICAL APPLICATIONS

34. The incentre of the triangle formed by the points $\bar{i}+\bar{j}+\bar{k}$, $4\bar{i}+\bar{j}+\bar{k}$ and $4\bar{i}+5\bar{j}+\bar{k}$ is

1) $\dfrac{\bar{i}+\bar{j}+\bar{k}}{3}$ 2) $\bar{i}+2\bar{j}+3\bar{k}$

3) $3\bar{i}+2\bar{j}+\bar{k}$ 4) $\bar{i}+\bar{j}+\bar{k}$

35. If the vectors $\bar{a}=3\bar{i}+\bar{j}-2\bar{k}$, $\bar{b}=-\bar{i}+3\bar{j}+4\bar{k}$, $\bar{c}=4\bar{i}-2\bar{j}-6\bar{k}$ form the sides of the triangle then length of the median bisecting the vector $\bar{c}$ is

1) $\sqrt{12}$ units 2) $\sqrt{6}$ units

3) $2\sqrt{6}$ units 4) $2\sqrt{3}$ units

36. If O is the circumcentre and O' is the orthocentre of a triangle ABC and if $\overline{AP}$ is the circumdiameter then

$\overline{AO'}+\overline{O'B}+\overline{O'C}=$

1) $\overline{OA}$ 2) $\overline{O'A}$ 3) $\overline{AP}$ 4) $\overline{AO}$

ANGULAR BISECTORS

37. Let O be the origin and A, B be two points. and $\bar{p},\bar{q}$ are vectors represented by $\overline{OA}$ and $\overline{OB}$ and their magnitudes are p,q. The unit vector bisecting the angle AOB is

1) $\dfrac{\frac{\bar{p}}{p}+\frac{\bar{q}}{q}}{\left|\frac{\bar{p}}{p}\right|+\left|\frac{\bar{q}}{q}\right|}$ 2) $\dfrac{\frac{\bar{p}}{p}+\frac{\bar{q}}{q}}{\left|\frac{\bar{p}}{p}\right|-\left|\frac{\bar{q}}{q}\right|}$ 3) $\dfrac{\frac{\bar{p}}{p}+\frac{\bar{q}}{q}}{\left|\frac{\bar{p}}{p}+\frac{\bar{q}}{q}\right|}$ 4) $\dfrac{\bar{p}+\bar{q}}{2}$

38. If $\bar{a}$ and $\bar{b}$ are two non-parallel unit vectors and the vector $\alpha\bar{a}+\bar{b}$ bisects the internal angle between $\bar{a}$ and $\bar{b}$, then α is

1) 1 2) 1/2 3) 2 4) 5

REGULAR HEXAGON

39. Let ABCDEF be a regular hexagon and $\overline{AB}=\bar{a}$, $\overline{BC}=\bar{b}$, $\overline{CD}=\bar{c}$ then $\overline{AE}$ is equal to

1) $\bar{a}+\bar{b}+\bar{c}$ 2) $\bar{b}+\bar{c}$

3) $\bar{a}+\bar{b}$ 4) $\bar{a}+\bar{c}$

40. Let ABCDEF be a regular hexagon. Then

$\overline{AB}+\overline{AC}+\overline{AD}+\overline{EA}+\overline{FA}=$

1) $2\overline{AB}$ 2) $3\overline{AB}$ 3) $4\overline{AB}$ 4) $\overline{AB}$

VECTOR EQUATION OF A LINE AND A PLANE

41. A point on the line

$\bar{r}=(1-t)\left(2\bar{i}+3\bar{j}+4\bar{k}\right)+t\left(3\bar{i}-2\bar{j}+2\bar{k}\right)$

1) $-\bar{i}+5\bar{j}+2\bar{k}$ 2) $\bar{i}+6\bar{j}+8\bar{k}$

3) $\bar{i}+8\bar{j}+6\bar{k}$ 4) $\bar{i}+\bar{j}+\bar{k}$

42. The point of intersection of

$\bar{r}=\bar{a}+s\left(\dfrac{\bar{b}+\bar{c}}{2}-\bar{a}\right)$, $\bar{r}=\bar{b}+t\left(\dfrac{\bar{c}+\bar{a}}{2}-\bar{b}\right)$

where $\bar{a},\bar{b},\bar{c}$, are position vectors of the vertices of a triangle

1) $\bar{a}+\bar{b}+\bar{c}$ 2) $\dfrac{\bar{a}+\bar{b}+\bar{c}}{2}$

3) $\dfrac{\bar{a}+\bar{b}+\bar{c}}{3}$ 4) $\dfrac{\bar{a}+\bar{b}+\bar{c}}{8}$

43. Cartesian equation of the plane $\bar{r}=(1+\lambda-\mu)\bar{i}+(2-\lambda)\bar{j}+(3-2\lambda+2\mu)\bar{k}$ is

1) $2x+y=5$ 2) $2x-y=5$

3) $2x+z=5$ 4) $2x-z=5$

LEVEL-I (C.W)-K E Y

01) 2	02) 2	03) 4	04) 3	05) 2	06) 3
07) 4	08) 2	09) 2	10) 3	11) 2	12) 1
13) 2	14) 1	15) 4	16) 3	17) 3	18) 2
19) 4	20) 4	21) 1	22) 2	23) 2	24) 3
25) 4	26) 4	27) 2	28) 3	29) 2	30) 4
31) 3	32) 3	33) 2	34) 3	35) 2	36) 3
37) 3	38) 1	39) 2	40) 3	41) 3	42) 3
43) 3					

LEVEL-I (C.W)-HINTS

1. $\sqrt{\cos^2\alpha\cos^2\beta+\cos^2\alpha\sin^2\beta+\sin^2\alpha}$

$=\sqrt{\cos^2\alpha+\sin^2\alpha}$

2. $\overline{AD}=\overline{AB}-\overline{CB}+\overline{CD}$

3. $\overline{AB}=\overline{OB}-\overline{OA}$ and, $\overline{AC}=\overline{OC}-\overline{OA}$

4. Required vector $=-\dfrac{\bar{a}}{|\bar{a}|}$

5. $\bar{a}+\bar{b}-\bar{c}=-2\bar{i}+\bar{j}-2\bar{k}$

$\left|\bar{a}+\bar{b}-\bar{c}\right|=3$, Required vector $=-\dfrac{\bar{a}+\bar{b}-\bar{c}}{\left|\bar{a}+\bar{b}-\bar{c}\right|}$

6. Two vectors are equal then corresponding components are equal.

7. $\frac{x}{-2}=\frac{-6}{3}=\frac{2}{y}$

8. $\left|\overline{PQ}\right|=\sqrt{9+27}=6$, $\left|\overline{RS}\right|=\sqrt{4+12}=4$

9. $\bar{b}=t\bar{a}\Rightarrow 21=|t||\bar{a}|$

10. $\frac{x}{1}=-\frac{2}{y}=-\frac{5}{3}$

11. Put $a=1$, $\left|\bar{p}+\bar{q}\right|=\left|\bar{p}\right|+\left|\bar{q}\right|$

12. $\begin{vmatrix}3 & 3 & \sqrt{3}\\ 1 & 0 & 1\\ \sqrt{3} & \sqrt{3} & \lambda\end{vmatrix}=0$

13. $\begin{vmatrix}x & -y & 0\\ x & 0 & -z\\ x-1 & -1 & -1\end{vmatrix}=0$

14. $\begin{vmatrix}1 & 1 & 1\\ 1 & -1 & 2\\ x & (x-2) & 1\end{vmatrix}=0$

$\Rightarrow x(2+1)-(x-2)\{2-1\}+1\{-1-1\}=0$

$\Rightarrow 3x-x+2-2=0 \Rightarrow 2x=0 \Rightarrow x=0$

15. $|\overline{BC}|=\sqrt{6},|\overline{CA}|=\sqrt{35},|\overline{AB}|=\sqrt{41}$

16. $\bar{a}=\bar{b}+\bar{c}$

17. $A=\bar{0},\overline{OD}=\frac{\bar{b}}{2},\overline{OE}=\frac{\bar{c}}{2}$

18. For an equilateral triangle, centroid = orthocentre.

19. $\overline{PQ}=\overline{AP}+\overline{PB}+\overline{PC}$,

$\Rightarrow \overline{CP}+\overline{PQ}=\overline{AP}+\overline{PB} \Rightarrow \overline{CQ}=\overline{AB}$

20. $\bar{a}=2\bar{i}+4\bar{j}-5\bar{k},\bar{b}=\bar{i}+2\bar{j}+3\bar{k}$

$\overline{AC}=\bar{a}+\bar{b}\quad \overline{BD}=\bar{b}-\bar{a}$

21. $2\left|\bar{i}+x\bar{j}+3\bar{k}\right|=\left|4\bar{i}+(4x-2)\bar{j}+2\bar{k}\right|$

$4\left(x^2+10\right)=20+\left(4x-2\right)^2$

$\Rightarrow 3x^2-4x-4=0$, $x=2,\frac{-2}{3}$

22. $\overline{PQ}=4\bar{i}-5\bar{j}+11\bar{k}$ direction cosines of

$\overline{PQ}=\frac{4}{\sqrt{162}}\bar{i}-\frac{5}{\sqrt{162}}\bar{j}+\frac{11}{\sqrt{162}}\bar{k}$

23. $\bar{a}=\lambda(6\bar{i}-8\bar{j}-\frac{15}{2}\bar{k})$;

$|\bar{a}|=50 \Rightarrow |\lambda|\sqrt{36+64+\frac{225}{4}}=50$

$\Rightarrow |\lambda|\left(\frac{25}{2}\right)=50 \Rightarrow |\lambda|=4 \quad \Rightarrow \lambda=\pm 4$

$\bar{a}=-4\bar{b}$ ($\because \bar{a}$ makes acute angle with z-axis)

24. C is mid point of AB

$\Rightarrow \overline{PC}=\frac{\overline{PA}+\overline{PB}}{2} \Rightarrow \overline{PA}+\overline{PB}=2\overline{PC}$

25. $\overline{AC}=2015\overline{AB}$, $\bar{c}-\bar{a}=2015\left(\bar{b}-\bar{a}\right)$

26. $\bar{c}=\frac{3\bar{a}+4\bar{b}}{7}$

27. $-(1+2):(1-7)=-3:-6=1:2$

28. $\overline{OC}=\frac{\overline{OA}+2\overline{OB}}{3}$

29. $\overline{OP}=\frac{3\overline{OB}-4\overline{OA}}{3-4}\quad \overline{OQ}=\frac{2\overline{OC}-\overline{OB}}{1}$

30. $\bar{i}+\bar{j}+\bar{k}$ is parallel to $\frac{1}{\sqrt{3}}\left(\bar{i}+\bar{j}+\bar{k}\right)$.

31. $l^2+m^2+n^2-lm-mn-nl\geq 0$

32. $2\bar{a}+3\bar{b}=\bar{c}$

33. $\begin{vmatrix}1 & 1343 & 1\\ -1 & 1 & 1\\ 1 & \mu & 2\end{vmatrix}=0$

34. $I=\frac{\overline{OA}\left|\overline{BC}\right|+\overline{OB}\left|\overline{CA}\right|+\overline{OC}\left|\overline{AB}\right|}{\left|\overline{BC}\right|+\left|\overline{CA}\right|+\left|\overline{AB}\right|}$

35. Length of the median through the vertex C=

$\frac{1}{2}\sqrt{2(a^2+b^2)-c^2}$

$=\frac{1}{2}\sqrt{2(14+26)-56}=\frac{1}{2}\sqrt{24}=\sqrt{6}$ units

36. $\overline{AO'}+\overline{O'B}+\overline{O'C}=2\overline{AO'}=\overline{AP}$

37. $\bar{r}=t\left(\frac{\bar{a}}{|\bar{a}|}+\frac{\bar{b}}{|\bar{b}|}\right)$

38. $\bar{r}=t\left(\frac{\bar{a}}{|\bar{a}|}+\frac{\bar{b}}{|\bar{b}|}\right)$

39. $\overline{AE}=\overline{AF}+\overline{FE}=\bar{b}+\bar{c}$

40. $\overline{FA}+\overline{AC}=\overline{FC}=2\overline{AB}$ and
$\overline{EA}+\overline{AD}=\overline{ED}=\overline{AB}$

41. If $t=-1$ then $\bar{r}=\bar{i}+8\bar{j}+6\bar{k}$

42. Point of intersection of medians is centroid

43. $\bar{r}=(\bar{i}+2\bar{j}+3\bar{k})+\lambda(\bar{i}-\bar{j}-2\bar{k})+\mu(-\bar{i}+2\bar{k})$
plane passing through A(1, 2, 3) and parallel to $\bar{b}=\bar{i}-\bar{j}-2\bar{k}$ and $\bar{c}=-\bar{i}+2\bar{k}$

LEVEL - I (H.W)

POSITION VECTOR,UNIT AND EQUAL VECTORS

1. $\bar{a}$ **is a non zero vector and** λ **is a scalar. Then** $\lambda\,\bar{a}$ **is a unit vector if**

1) $\lambda=|\bar{a}|$ 2) $\lambda=\pm\frac{1}{|\bar{a}|}$ 3) $\lambda=\frac{4}{|\bar{a}|}$ 4) $\lambda=\frac{2}{|\bar{a}|}$

2. **Let** $\overline{OA}=2\bar{i}+4\bar{k}$**,**
$\overline{OB}=5\bar{i}+3\sqrt{3}\bar{j}+4\bar{k}$, $\overline{OC}=-2\sqrt{3}\bar{j}+\bar{k}$,
$\overline{OD}=2\bar{i}+\bar{k}$ **.Then the value of** $\overline{CD}$ **in terms of** $\overline{AB}$ **is**

1) $\overline{AB}$ 2) $\frac{1}{3}\overline{AB}$ 3) $\frac{2}{3}\overline{AB}$ 4) $\frac{1}{4}\overline{AB}$

3. **If the vectors** $\bar{a}=x\bar{i}+2\bar{j}+z\bar{k}$ **and** $\bar{b}=2\bar{i}+y\bar{j}+\bar{k}$ **are equal, then** $(x,y,z)=$

1) (2, 1, 2) 2) (2, 2, 1) 3) (1, 2, 2) 4) (2, 1, 1)

4. **The unit vector in the direction of** $2\bar{i}+3\bar{j}+\bar{k}$ **is**

1) $\pm\frac{1}{\sqrt{14}}(2\bar{i}+3\bar{j}+\bar{k})$ 2) $-\frac{1}{\sqrt{14}}(2\bar{i}+3\bar{j}+\bar{k})$

3) $\frac{1}{\sqrt{14}}(2\bar{i}+3\bar{j}+\bar{k})$ 4) $\sqrt{14}(2\bar{i}+3\bar{j}+\bar{k})$

5. **If** $\bar{a}=3\bar{i}-2\bar{j}+\bar{k}$ **and** $\bar{b}=\bar{i}+2\bar{j}+5\bar{k}$ **then unit vector in the direction of** $\bar{a}-\bar{b}$ **is**

1) $\frac{1}{\sqrt{34}}(2\bar{i}-4\bar{j}+4\bar{k})$ 2) $\frac{1}{3}(\bar{i}-2\bar{j}-2\bar{k})$

3) $\frac{1}{6}(-2\bar{i}-4\bar{j}-2\bar{k})$ 4) $\frac{1}{4}(\bar{i}-2\bar{j}-2\bar{k})$

COLLINEAR,LIKE AND UNLIKE VECTORS

6. **The position vectors of the points A,B, C are respectively** $\bar{i}+x\bar{j}+3\bar{k}$**,** $3\bar{i}+4\bar{j}+7\bar{k}$ **and** $y\bar{i}-2\bar{j}-5\bar{k}$**. If A,B, and C are collinear, then** $|x\bar{i}+y\bar{j}|=$

1) $\sqrt{13}$ 2) $\sqrt{11}$ 3) 3 4) $2\sqrt{2}$

7. **Three points whose position vectors are** $x\bar{i}+y\bar{j}+z\bar{k}$, $\bar{i}+z\bar{j}$ **and** $-\bar{i}-\bar{j}$ **are collinear, then relation between x, y, z is**

1) $x-2y=1, z=0$ 2) $x+y=1, z=0$

3) $x-y=1, z=0$ 4) $x+2y=1, z=0$

8. **The position vectors of four points P,Q,R,S are** $2\bar{a}+4\bar{c}$, $5\bar{a}+3\sqrt{3}\bar{b}+4\bar{c}$**,** $-2\sqrt{3}\bar{b}+\bar{c}$ **and** $2\bar{a}+\bar{c}$ **respectively, then**

1) $\overline{PQ}$ is parallel to $\overline{RS}$

2) $\overline{PQ}$ is not parallel to $\overline{RS}$

3) $\overline{PQ}$ is equal to $\overline{RS}$

4) $\overline{PQ}$ is parallel and equal to $\overline{RS}$

9. **If the vectors** $2\bar{i}-3\bar{j}+6\bar{k}$, $\bar{b}$ **are collinear and** $|\bar{b}|=14$ **then** $\bar{b}=$

1) $\frac{1}{7}(-2\bar{i}+3\bar{j}+6\bar{k})$ 2) $\pm(4\bar{i}-6\bar{j}+12\bar{k})$

3) $6\bar{i}-9\bar{j}+18\bar{k}$ 4) $\frac{1}{7}(2\bar{i}-3\bar{j}+6\bar{k})$

10. **The three points A,B,C have position vectors** *(1,x,3),(3,4,7)* **and** *(y,-2,-5)* **are collinear then** *(x,y)* **(EAM-2002)**

1) (2,-3) 2) (-2,3) 3) (-2,-3) 4) (2,3)

11. Let $\bar{a}=\bar{i}+\bar{j}-\bar{k}$, $\bar{b}=5\bar{i}-3\bar{j}-3\bar{k}$, $\bar{c}=3\bar{i}-\bar{j}+2\bar{k}$. If a vector $\bar{r}$ is collinear with $\bar{c}$ and $|\bar{r}|=|\bar{a}+\bar{b}|$, then $\bar{r}$ equals

1) $\pm3\bar{c}$ 2) $\pm\frac{3}{2}\bar{c}$ 3) $\pm2\bar{c}$ 4) $\pm\frac{2}{3}\bar{c}$

COPLANAR AND NON COPLANAR VECTORS

12. Let $\bar{a}=\bar{i}+\bar{j}$, $\bar{b}=\bar{j}+\bar{k}$ and $\bar{c}=\alpha\bar{a}+\beta\bar{b}$. If the vectors $\bar{i}-2\bar{j}+\bar{k}$, $3\bar{i}+2\bar{j}-\bar{k}$ and $\bar{c}$ are coplanar then $\frac{\alpha}{\beta}=$

1) 1 2) 2 3) 3 4) -3

13. If the vectors $2\bar{i}+3\bar{j}-6\bar{k}, 6\bar{i}-2\bar{j}+3\bar{k}$, $3\bar{i}-6\bar{j}-2\bar{k}$ and $\bar{i}+\bar{j}-\lambda^2\bar{k}$ are coplanar then $31\lambda^2-233$ is

1) 0 2) 2 3) 1 4) 3

14. Let a,b and c be distinct non-negative numbers. If the vectors $a\bar{i}+a\bar{j}+c\bar{k}$, $\bar{i}+\bar{k}$ and $c\bar{i}+c\bar{j}+b\bar{k}$ lie in a plane , then c is [AIE-2005]

1) The harmonic mean of a and b
2) equal to zero
3) the arithmetic mean of a and b
4) the geometric mean of a and b

TRIANGLE AND PARALLELOGRAM LAW, VECTOR ADDITION

15. The points A(–1, 6, 6) B(–4, 9, 6) C(0, 7, 10) form

1) equilateral triangle 2) right angled triangle
3) right angled isosceles triangle
4) isosceles triangle

16. If $2\bar{i}+3\bar{j}-6\bar{k}, 6\bar{i}-2\bar{j}+3\bar{k}, 3\bar{i}-6\bar{j}-2\bar{k}$ represent the sides of a triangle,then the perimeter of the triangle is

1) 6 2) 7 3) 14 4) 21

17. Let ABCD is a parallelogram and $\overline{AC}$, $\overline{BD}$ be its diagonals Then $\overline{AC}+\overline{BD}$ is

1) $2\overline{AB}$ 2) $2\overline{BC}$ 3) $3\overline{AB}$ 4) $3\overline{BC}$

18. If G is the centroid of ΔABC, $\overline{GA}+\overline{BG}+\overline{GC}=$

1) $2\overline{GB}$ 2) $2\overline{GA}$ 3) $\bar{O}$ 4) $2\overline{BG}$

19. The vectors $\bar{a},\bar{b},\bar{c},\bar{d}$ are respectively the p.v's of the points A, B, C, D and P, Q, R, S are the points such that
$\overline{AP}=2\overline{PB}$; $\overline{BQ}=2\overline{QC}$;
$\overline{CR}=2\overline{RD}$; $\overline{DS}=2\overline{SA}$. If PQRS is a parallelogram, then $\bar{a}-\bar{b}+\bar{c}-\bar{d}=$

1) $\bar{0}$ 2) 0 3) $3\bar{a}$ 4) $2\bar{a}$

20. If $\bar{i}+2\bar{j}+3\bar{k}$, $3\bar{i}+2\bar{j}+\bar{k}$, are sides of a parallelogram, then a unit vector parallel to one of the diagonals is

1) $\frac{1}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})$ 2) $\frac{1}{\sqrt{3}}(\bar{i}-\bar{j}+\bar{k})$

3) $\frac{1}{\sqrt{3}}(\bar{i}+\bar{j}-\bar{k})$ 4) $\frac{1}{\sqrt{3}}(-\bar{i}+\bar{j}+\bar{k})$

SCALAR MULTIPLICATION AND THEIR PROPERTIES

21. The vector $\bar{i}+2x\bar{j}+2\bar{k}$ rotated through an angle θ and doubled in magnitude then it becomes $2\bar{i}+(2x+2)\bar{j}+(6x-2)\bar{k}$. Then the values of x are

1) 1, 1/3 2) -1, 1/3 3) 1,-1/3 4) 0,3

ANGLE BETWEEN TWO VECTORS

22. Unit vector making angles $\frac{\pi}{6},\frac{\pi}{6},\frac{\pi}{3}$ with $\bar{i},\bar{j},\bar{k}$ directions is

1) $\frac{1}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})$ 2) $\frac{1}{\sqrt{3}}(\bar{i}-\bar{j}+\bar{k})$

3) $\frac{1}{\sqrt{3}}(\bar{i}-\bar{j}-\bar{k})$

4) impossible to get such a vector

23. The non zero vectors $\bar{a},\bar{b}$ and $\bar{c}$ are related by $\bar{a}=8\bar{b}$ and $\bar{c}=-7\bar{b}$ angle between $\bar{a}$ and $\bar{c}$ is [AIE-2007]

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) π 4) 0

SECTION FORMULA,MID POINT

24. If $\overline{AO}+\overline{OB}=\overline{BO}+\overline{OC}$ then

1) A is the midpoint of $\overline{BC}$
2) B is the midpoint of $\overline{CA}$
3) C is the midpoint of $\overline{AB}$
4) C divides $\overline{AB}$ in the ratio 1:2

25. The position vectors of P and Q are respectively $\bar{a}$ and $\bar{b}$. If R is a point on $\overline{PQ}$ such that $\overline{PR} = 5\overline{PQ}$, then the position vector of R is

1) $5\bar{b}-4\bar{a}$ 2) $5\bar{b}+4\bar{a}$ 3) $4\bar{b}-5\bar{a}$ 4) $4\bar{b}+5\bar{a}$

26. If A, B, C are collinear points whose position vectors are $\bar{a},\bar{b},\bar{c}$ respectively satisfying the condition $3\bar{a}+2\bar{c}=5\bar{b}$, then AB : BC =

1) 3 : 2 2) 5 : 3 3) 2 : 3 4) 3 : 5.

27. Let A =(-3,4,-8) ,B=(5,-6,4). Then the coordinates of the point in which the XY- plane or XOY plane divides the line segment $\overline{AB}$ is

1) (7,-8,0) 2) $\left(\frac{7}{3},-\frac{8}{3},0\right)$

3) $\left(\frac{7}{2},-\frac{8}{2},0\right)$ 4) $\left(0,\frac{7}{3},-\frac{8}{3}\right)$

28. Let $A=2\bar{i}+7\bar{j}$, $B=\bar{i}+2\bar{j}+4\bar{k}$, $C=\dfrac{9\bar{i}+30\bar{j}+4\bar{k}}{5}$. Then the ratio in which C divides $\overline{AB}$ internally is

1) 1 : 4 2) 2 : 3 3) 3 : 2 4) 5 : 1

29. Let $A(\bar{a}),B(\bar{b}),C(\bar{c})$ be the vertices of the triangle ABC and let D,E,F be the mid points of the sides BC,CA,AB respectively.If P divides the median AD in the ratio 2:1 then the position vector of P is

1) $\bar{0}$ 2) $\bar{a}+\bar{b}+\bar{c}$

3) $\dfrac{\bar{a}+\bar{b}+\bar{c}}{3}$ 4) $\dfrac{2\bar{a}+\bar{b}+\bar{c}}{3}$

LINEAR COMBINATION,LINEARLY INDEPENDENT AND DEPENDENT VECTORS,DC'S & DR'S

30. A straight line is inclined to the axes of Y and Z at angles 45^0 *and* 60^0 respectively. The inclination of the line with the X-axis is

1) 60^0 2) 45^0 3) 30^0 4) 90^0

31. If $\bar{e}=l\bar{i}+m\bar{j}+n\bar{k}$ is a unit vector and $l=\dfrac{1}{3}$,then maximum value of lmn is

1)1 2) $\dfrac{4}{27}$ 3) $\dfrac{8}{27}$ 4) $\dfrac{3}{2}$

32. If $(x,y,z)\neq 0$ and $(\bar{i}+\bar{j}+3\bar{k})x$ $+(3\bar{i}-3\bar{j}+\bar{k})y$ $+(-4\bar{i}+5\bar{j})z$ $=\alpha(x\bar{i}+y\bar{j}+z\bar{k})$, then $\alpha=$

1) 0, 1 2) −1, 0 3) 0, 2 4) −2, 0

33. If the vectors $\bar{a}+\bar{b}+\bar{c}$, $\bar{a}+\lambda\bar{b}+2\bar{c}$, $-\bar{a}+\bar{b}+\bar{c}$ are linearly dependent then λ =

1) 1 2) 2 3) 3 4) 4

GEOMETRICAL APPLICATIONS

34. The position vectors of A,B,C are $\bar{i}+\bar{j}+\bar{k},4\bar{i}+\bar{j}+\bar{k},4\bar{i}+5\bar{j}+\bar{k}$ Then the position vector of the circumcentre of the triangle ABC is

1) $3\bar{i}+2\bar{j}+\bar{k}$ 2) $\frac{1}{2}(6\bar{i}+\bar{j}+\bar{k})$

3) $\frac{1}{2}(5\bar{i}+6\bar{j}+2\bar{k})$ 4) $\frac{1}{2}(9\bar{i}+7\bar{j}+3\bar{k})$

35. The vectors $\overline{AB}=3\bar{i}+4\bar{k}$ and $\overline{AC}=5\bar{i}-2\bar{j}+4\bar{k}$ are the sides of a triangle ABC. The length of the median through A is [EAM-2011][MAINS-2013]

1) $\sqrt{72}$ 2) $\sqrt{33}$ 3) $\sqrt{288}$ 4) $\sqrt{18}$

36. Let G and $G^{|}$ be the centroids of the triangles ABC and $A^{|}B^{|}C^{|}$ respectively.Then $\overline{AA'}+\overline{BB'}+\overline{CC'}=$

1) $2\overline{GG^1}$ 2) $3\overline{G^1G}$ 3) $3\overline{GG^1}$ 4) $\frac{3}{2}\overline{GG^1}$

ANGULAR BISECTORS

37. The unit vector bisecting $\overline{OY}$ and $\overline{OZ}$ is

1) $\dfrac{\bar{i}+\bar{j}+\bar{k}}{\sqrt{3}}$ 2) $\dfrac{\bar{j}-\bar{k}}{\sqrt{2}}$ 3) $\dfrac{\bar{j}+\bar{k}}{\sqrt{2}}$ 4) $\dfrac{-\bar{j}+\bar{k}}{\sqrt{2}}$

38. The vector $a\bar{i}+b\bar{j}+c\bar{k}$ is a bisector of the angle between the vectors $\bar{i}+\bar{j}$ and $\bar{j}+\bar{k}$ if

1) $a=b$ 2) $a=c$ 3) $a+b=c$ 4) $a=b=c$

REGULAR HEXAGON

39. Let ABCDEF be a regular hexagon and let $\overline{AB}=\bar{a}$, $\overline{BC}=\bar{b}$, then $\overline{CD}$=

1) $\bar{a}-\bar{b}$ 2) $\bar{a}+\bar{b}$ 3) $\bar{b}-\bar{a}$ 4) $2\bar{a}$

40. **Let ABCDEF be a regular hexagon, If $\overline{AD} = x\overline{BC}$ and $\overline{CF} = y\overline{BA}$ then $(x+y)^2+8=$**

1) 24 2) -4 3) 2 4) -24

VECTOR EQUATION OF A LINE AND A PLANE:

41. **A point on the line $\overline{r} = 2\overline{i} + 3\overline{j} + 4\overline{k} + t(\overline{i} + \overline{j} + \overline{k})$ is**

1) (2014,2015,2016) 2) (2013,2015,2017)
3) (2013,2014,2017) 4) (0,0,0)

42. **The point of intersection of the lines $\overline{r} = \overline{a} + s(\overline{b} + \overline{c})$, $\overline{r} = \overline{b} + t(\overline{a} + \overline{c})$ (where $\overline{a}, \overline{b}, \overline{c}$ are L.I.) is**

1) $\overline{0}$ 2) $\overline{a}+\overline{b}$ 3) $\overline{a}+\overline{b}+\overline{c}$ 4) $\overline{b}+\overline{c}$

43. **The vector equation of the plane through the points (1,-2,-3) and parallel to the vectors (2,-1,3) and (2,3,-6) is $\overline{r}=$**

1) $(1+2t+2s)\overline{i} - (2+t-3s)\overline{j} - (3-3t+6s)\overline{k}$

2) $(1+2t+2s)\overline{i} + (2+t+3s)\overline{j} - (3+3t+6s)\overline{k}$

3) $(1+2t+2s)\overline{i} + (2+t+3s)\overline{j} + (3+3t+6s)\overline{k}$

4) $(1+2t+2s)\overline{i} + (2+t-3s)\overline{j} + (3+3t+6s)\overline{k}$

LEVEL-I (H.W)-K E Y

01) 2	02) 3	03) 2	04) 3	05) 2	06) 1
07) 1	08) 1	09) 2	10) 1	11) 3	12) 4
13) 1	14) 4	15) 3	16) 4	17) 2	18) 4
19) 1	20) 1	21) 3	22) 4	23) 3	24) 2
25) 1	26) 3	27) 2	28) 1	29) 3	30) 1
31) 2	32) 2	33) 2	34) 3	35) 2	36) 3
37) 3	38) 2	39) 3	40) 1	41) 1	42) 3
43) 1					

LEVEL-I (H.W)-HINTS

1. Given, $|\lambda \overline{a}| = 1 \Rightarrow \lambda = \pm\frac{1}{|\overline{a}|}$

2. $\overline{AB} = 3\overline{i} + 3\sqrt{3}\overline{j}$, $\overline{CD} = 2\overline{i} + 2\sqrt{3}\overline{j}$

3. Two vectors are equal then corresponding components are equal.

4. Unit vector in the direction of $\overline{a} = \frac{\overline{a}}{|\overline{a}|}$.

5. Required vector $= \frac{\overline{a}-\overline{b}}{|\overline{a}-\overline{b}|}$

6. A,B,C are collinear $\Rightarrow \overline{AB} = \lambda\overline{BC}$

$\Rightarrow \lambda = -\frac{1}{3}, x = 4+6\lambda = 2, y = 3-6 = -3$

Therefore $|x\overline{i} + y\overline{j}| = \sqrt{2^2+(-3)^2} = \sqrt{13}$

7. $\overline{OA} = (x,y,z), \overline{OB} = (1,z,0), \overline{OC} = (-1,-1,0)$

$\overline{AB} = (1-x, z-y, -z)$

$\overline{AC} = (-1+x, -(1+y), -z)$

$\overline{AB} = \lambda\overline{AC} \Rightarrow 1-x = -\lambda(1+x)$

$z-y = -\lambda(1+y)$, $-z = -\lambda z$

$\lambda \neq 0, \Rightarrow z = 0 \Rightarrow 2y = x-1$

8. $\overline{PQ} = 3(\overline{a} + \sqrt{3}\overline{b})$ and $\overline{RS} = 2(\overline{a} + \sqrt{3}\overline{b})$

$\overline{PQ}$ is parallel to $\overline{RS}$

9. $\overline{b} = t(2\overline{i} - 3\overline{j} + 6\overline{k}) \Rightarrow |\overline{b}| = |t|7$

10. $x_1 - x_2 : x_2 - x_3 = y_1 - y_2 : y_2 - y_3$

$= z_1 - z_2 : z_2 - z_3$

11. $\overline{a} + \overline{b} = 6\overline{i} - 2\overline{j} - 4\overline{k}$ *so that*

$|\overline{a}+\overline{b}| = \sqrt{56} \Rightarrow \sqrt{56} = |\lambda|\sqrt{14}$

12. We have $\overline{c} = \alpha(\overline{i}+\overline{j}) + \beta(\overline{j}+\overline{k})$

$$\begin{vmatrix} 1 & -2 & 1 \\ 3 & 2 & -1 \\ \alpha & \alpha+\beta & \beta \end{vmatrix} = 0 \Rightarrow \frac{\alpha}{\beta} = -3$$

13. $$\begin{vmatrix} 4 & -5 & 9 \\ 1 & -9 & 4 \\ -1 & -2 & 6-\lambda^2 \end{vmatrix} = 0$$

14. Since, the given vectors lie in a plane,

$$\therefore \begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0 \Rightarrow c^2 = ab$$

15. Find $|\overline{AB}|, |\overline{BC}|, |\overline{CA}|$

16. AB=BC=CA=7 $\therefore$ perimeter $=21$

17. $\overline{AC} = \overline{a} + \overline{b}$, $\overline{BD} = \overline{b} - \overline{a}$

18. Use $\overline{GA} + \overline{GB} + \overline{GC} = \overline{0}$

19. $3\overline{OP} = \overline{OA} + 2\overline{OB}; 3\overline{OQ} = \overline{OB} + 2\overline{OC}$

$3\overline{OR} = \overline{OC} + 2\overline{OD}$, $3\overline{OS} = \overline{OD} + 2\overline{OA}$ and

$\frac{\overline{OP}+\overline{OR}}{2} = \frac{\overline{OQ}+\overline{OS}}{2}$

20. $\overline{AC}=4\bar{i}+4\bar{j}+4\bar{k}$, $\frac{\overline{AC}}{|\overline{AC}|}=\frac{\bar{i}+\bar{j}+\bar{k}}{\sqrt{3}}$

21. $2|\bar{i}+2x\bar{j}+2\bar{k}|=|2\bar{i}+(2x+2)\bar{j}+(6x-2)\bar{k}|$

$\Rightarrow 4\sqrt{1+4x^2+4}=\sqrt{4+(2x+2)^2+(6x-2)^2}$

$\Rightarrow 3x^2-2x-1=0$

22. $\cos^2\frac{\pi}{6}+\cos^2\frac{\pi}{6}+\cos^2\frac{\pi}{3}\neq 1$

23. $\bar{a},\bar{b}$ vectors are in same direction. $\bar{b}$ and $\bar{c}$ are in opposite direction $\Rightarrow \bar{a}$ and $\bar{c}$ are in opposite directions.

24. $\overline{AB}=\overline{BC}\Rightarrow B$ is the mid point of $\overline{AC}$

25. Given $\overline{PR}=5\overline{PQ}$. It mean R divides PQ externally in the ratio 5 : 4

$\therefore$ Position vector of $R=\frac{5\bar{b}-4\bar{a}}{5-4}=5\bar{b}-4\bar{a}$

26. $3\bar{a}+2\bar{c}=5\bar{b}\Rightarrow\frac{3\bar{a}+2\bar{c}}{5}=\bar{b}$

$\Rightarrow$ B divides AC in the ratio 2 : 3.

27. XY plane devides AB in the ratio $-z_1:z_2$

28. $x_1-x_3:x_3-x_2=2-\frac{9}{5}:\frac{9}{5}-1$

29. $\overline{OP}=\frac{2\overline{OD}+\overline{OA}}{3}$ or P is the centroid

30. $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

31. $l^2+m^2+n^2=1$ and $l=\frac{1}{3}$, for maxima m = n

32. Equating the components of $\bar{i},\bar{j},\bar{k}$

$$\Rightarrow\begin{vmatrix}1-\alpha & 3 & -4\\ 1 & -(\alpha+3) & 5\\ 3 & 1 & -\alpha\end{vmatrix}=0$$

33. $\begin{vmatrix}1 & 1 & 1\\ 1 & \lambda & 2\\ -1 & 1 & 1\end{vmatrix}=0$

34. $S=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2},z_1\right)$

35. Median $\overline{AD}=\frac{\overline{AB}+\overline{AC}}{2}$

36. $\overline{AA'}+\overline{BB'}+\overline{CC'}=$

$\left(\overline{OA'}+\overline{OB'}+\overline{OC'}\right)-\left(\overline{OA}+\overline{OB}+\overline{OC}\right)$

37. $\bar{j},\bar{k}$ are unit vectors along Y and Z axes then unit vector bisecting $\overline{OY}$ and $\overline{OZ}$ is $\frac{\bar{j}+\bar{k}}{\sqrt{2}}$

38. $\bar{r}=t\left(\frac{\bar{a}}{|\bar{a}|}+\frac{\bar{b}}{|\bar{b}|}\right)$

39. $\overline{CD}=\bar{b}-\bar{a}$

40. $\overline{AD}=2\overline{BC}\Rightarrow x=2$

$\overline{CF}=2\overline{BA}\Rightarrow y=2\quad\Rightarrow(x+y)^2+8=24$

41. Cartesian equation is

$\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{1}=t$

42. $\bar{a}+s\left(\bar{b}+\bar{c}\right)=\bar{b}+t\left(\bar{c}+\bar{a}\right)$

43. $\bar{r}=\bar{a}+t\bar{b}+s\bar{c}$

LEVEL - II (C.W)

POSITION VECTOR,UNIT AND EQUAL VECTORS:

1. $p\bar{i}+3\bar{j}+4\bar{k}$ and $\sqrt{q}\,\bar{i}+5\bar{k}$ are two vectors where $p,q\geq 0$ are two scalars. Then the lengths of the vector are equal for

1) all values of (p,q)

2) only finite number of values of (p,q)

3) infinite number of values of (p,q)

4) No value of (p,q)

COLLINEAR,LIKE AND UNLIKE VECTORS:

2. If the points $\bar{a}+\bar{b},\bar{a}-\bar{b},\bar{a}+k\bar{b}$ are collinear, then

1) k has only one real value

2) k has two real value 3) k has all real values

4) k has finite number of real values

3. Let $\bar{f}(t)=[t]\bar{i}-(t-[t])\bar{j}+[t+1]\bar{k}$ be a vector. Where $[.]$ is a greatest integer function. If $\bar{f}\left(\frac{5}{4}\right)$ and $\bar{i}+\lambda\bar{j}+\mu\bar{k}$ are parallel vectors then $(\lambda,\mu)=$

1) (1, 1) 2) $\left(\frac{-1}{4},2\right)$ 3) $\left(\frac{1}{2},2\right)$ 4) $\left(\frac{1}{4},4\right)$

4. The positions vector of three points are $2\bar{a}-\bar{b}+3\bar{c}$, $\bar{a}-2\bar{b}+x\bar{c}$ and $y\bar{a}-5\bar{b}$ where $\bar{a},\bar{b},\bar{c}$ are non-coplanar vectors. If the three points are collinear, then

1) $x=-2, y=\frac{9}{4}$ 2) $x=\frac{-9}{4}, y=2$

3) $x=\frac{9}{4}, y=-2$ 4) $x=-2, y=-2$

5. If $\bar{r}=3\bar{p}+4\bar{q}$ and $2\bar{r}=\bar{p}-3\bar{q}$ then

1) $\bar{r},\bar{q}$ have same direction and $|\bar{r}|<2|\bar{q}|$

2) $\bar{r},\bar{q}$ have opposite direction and $|\bar{r}|>2|\bar{q}|$

3) $\bar{r},\bar{q}$ have opposite direction and $|\bar{r}|<2|\bar{q}|$

4) $\bar{r},\bar{q}$ have same direction and $|\bar{r}|>2|\bar{q}|$

COPLANAR AND NON COPLANAR VECTORS

6. If $\bar{a},\bar{b},\bar{c}$ are non coplanar vectors and λ is a real number, then the vectors $\bar{a}+2\bar{b}+3\bar{c}, \lambda\bar{b}+4\bar{c}$ and $(2\lambda-1)\bar{c}$ are non coplanar for (AIE-2004)

1) all except two values of λ

2) all except one value of λ

3) for all values of λ 4) no value of λ

TRIANGLE AND PARALLELOGRAM LAW, VECTOR ADDITION:

7. The value of 'a', for which the points A, B, C with position vectors $2\bar{i}-\bar{j}+\bar{k}, \bar{i}-3\bar{j}-5\bar{k}$ and $a\bar{i}-3\bar{j}+\bar{k}$ are the vertices of a right angled triangle with $\underline{|C}=\frac{\pi}{2}$ are [AIE-2006]

1) –2 and 1 2) 2 and –1

3) 2 and 1 4) –2 and –1

SCALAR MULTIPLICATION AND THEIR PROPERTIES

8. Let A and B be points with position vectors $\bar{a}$ and $\bar{b}$ with respect to origin O. If the point 'C' on OA is such that $2\overline{AC}=\overline{CO}, \overline{CD}$ is parallel to $\overline{OB}$ and $|\overline{CD}|=3|\overline{OB}|$ then $\overline{AD}$ is

1) $\bar{b}-\frac{\bar{a}}{9}$ 2) $3\bar{b}-\frac{\bar{a}}{3}$ 3) $\bar{b}-\frac{\bar{a}}{3}$ 4) $\bar{b}+\frac{\bar{a}}{3}$

SECTION FORMULA,MID POINT

9. A point $C=\frac{5\bar{a}+4\bar{b}-5\bar{c}}{3}$ divides the line joining the points $A=\bar{a}-2\bar{b}+3\bar{c}$ and B in the ratio 2:1, then the position vector of B is

1) $\bar{a}+3\bar{b}-4\bar{c}$ 2) $2\bar{a}-3\bar{b}+4\bar{c}$

3) $2\bar{a}+3\bar{b}+4\bar{c}$ 4) $2\bar{a}+3\bar{b}-4\bar{c}$

10. In quadrilateral ABCD , $\overline{AB}=\bar{a}, \overline{BC}=\bar{b}, \overline{AD}=\bar{b}-\bar{a}$. If M is the mid point of BC and N is a point on DM such that $DN=\frac{4}{5}DM$, then $\overline{AN}=$

1) $\frac{1}{5}\overline{AC}$ 2) $\frac{2}{5}\overline{AC}$ 3) $\frac{3}{5}\overline{AC}$ 4) $\frac{4}{5}\overline{AC}$

11. The median AD of ΔABC is bisected at E and BE is produced to meet the side AC in F. Then the ratio AF : FC =

1) 1 : 3 2) 2 : 1 3) 1 : 2 4) 3 : 1

LINEAR COMBINATION,LINEARLY INDEPENDENT AND DEPENDENT VECTORS, DC'S & DR'S:

12. If $\bar{r}=3\bar{i}+2\bar{j}-5\bar{k}$,$\bar{a}=2\bar{i}-\bar{j}+\bar{k}$, $\bar{b}=\bar{i}+3\bar{j}-2\bar{k}$ and $\bar{c}=-2\bar{i}+\bar{j}-3\bar{k}$ such that $\bar{r}=\lambda\bar{a}+\mu\bar{b}+\nu\bar{c}$. Then

1) $\mu, \frac{\lambda}{2}, \nu$ are in A.P. 2) λ, μ, ν are in A.P

3) λ, μ, ν are in H.P. 4) λ, μ, ν are in G.P.

13. If $\overline{OA}=3\bar{i}+\bar{j}-\bar{k}$, $|\overline{AB}|=2\sqrt{6}$ and AB has the direction ratios 1,-1,2 then |OB| =

1) $\sqrt{35}$ 2) $\sqrt{41}$ 3) $\sqrt{26}$ 4) $\sqrt{55}$

14. A vector $\bar{e}$ whose magnitude is 10 and making equal angles less than 90^0 with the coordinate axes is

1) $\frac{1}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})$ 2) $\frac{10}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})$

3) $\frac{10}{\sqrt{3}}(\bar{i}-\bar{j}+\bar{k})$ 4) $\frac{10}{\sqrt{3}}(\bar{i}-\bar{j}-\bar{k})$

GEOMETRICAL APPLICATIONS

15. A vector $\bar{a}$ has components a_1, a_2, a_3 in a right handed rectangular cartesian coordinate system OXYZ .The coordinate system is rotated about z-axis through an angle $\pi/2$ .The components of $\bar{a}$ in the new system are

1) $(-a_1, a_2, a_3)$ 2) (a_2, a_1, a_3)
3) $(a_2, -a_1, a_3)$ 4) $(a_2, a_1, -a_3)$

16. **Given three vectors $\bar{a} = 6\bar{i} - 3\bar{j}, \bar{b} = 2\bar{i} - 6\bar{j}$ and $\bar{c} = -2\bar{i} + 21\bar{j}$ such that $\bar{\alpha} = \bar{a} + \bar{b} + \bar{c}$ .Then the resolution of the vector $\bar{\alpha}$ into components with respect to $\bar{a}$ and $\bar{b}$ given by**

1) $3\bar{a} - 2\bar{b}$ 2) $3\bar{b} - 2\bar{a}$ 3) $2\bar{a} - 3\bar{b}$ 4) $\bar{a} - 3\bar{b}$

17. If $\bar{a}, \bar{b}, \bar{c}$ are position vectors of vertices A, B, C of ΔABC . If $\bar{r}$ is position vector of a point P such that $\left(|\bar{b} - \bar{c}| + |\bar{c} - \bar{a}| + |\bar{a} - \bar{b}|\right)\bar{r}$ $= |\bar{b} - \bar{c}|\bar{a} + |\bar{c} - \bar{a}|\bar{b} + |\bar{a} - \bar{b}|\bar{c}$ then the point P is

1) centroid of ΔABC 2) Orthocentre of ΔABC
3) circumcentre of ΔABC 4) incentre of ΔABC

18. The position vectors of the vertices of a triangle are $3\bar{i}+4\bar{j}+5\bar{k}, \bar{i}+7\bar{k}$ and $5\bar{i}+5\bar{j}$ The distance between the circumcentre and orthocentre is

1) 0 2) $\sqrt{306}$ 3) $2\sqrt{306}$ 4) $\frac{3}{2}\sqrt{306}$

19. The position vectors A, B are $\bar{a}, \bar{b}$ respectively. The position vector of C is $\frac{5\bar{a}}{3} - \bar{b}$. Then

1) C is outside the ΔOAB but inside the angle OBA
2) C is outside the ΔOAB but inside the angle BOA
3) C is outside the ΔOAB but inside the angle COA
4) inside the triangle OAB

20. If (2, -1, 2) is the centroid of tetrahedron OABC and G_1 is the centroid of ΔABC then $|\overline{OG_1}| =$

1) 4 2) 1 3) 9/2 4) 3/2

ANGULAR BISECTORS

21. If $4\bar{i} + 7\bar{j} + 8\bar{k}$, $2\bar{i} + 3\bar{j} + 4\bar{k}$ and $2\bar{i} + 5\bar{j} + 7\bar{k}$ are the position vectors of the vertices A,B and C of triangle ABC, the position vector of the point where the bisector of $\angle A$ meets BC is

1) $\frac{2}{3}(-6\bar{i} - 8\bar{j} - 6\bar{k})$ 2) $\frac{2}{3}(6\bar{i} + 8\bar{j} + 6\bar{k})$

3) $\frac{1}{3}(6\bar{i} + 13\bar{j} + 18\bar{k})$ 4) $2(\bar{i} + \bar{j} + \bar{k})$

22. Let A (4,7,8) B (2,3,4) and C (2,5,7) be the position vectors of the vertices of a Δ ABC. Then the length of internal angular bisector of angle A is

1) $\frac{3}{2}\sqrt{34}$ 2) $\frac{2}{3}\sqrt{34}$ 3) $\frac{1}{2}\sqrt{34}$ 4) $\frac{1}{3}\sqrt{34}$

REGULAR POLYGON

23. **If $A_1A_2......A_n$ is a regular polygon. Then the vector $\overline{A_1A_2} + \overline{A_2A_3} + ... + \overline{A_nA_1}$ is equal to**

1) $\bar{0}$ 2) $n\left(\overline{A_1A_2}\right)$

3) $n\left(\overline{OA_1}\right)$ (O is the centre) 4) $(n-1)\left(\overline{A_1A_2}\right)$

VECTOR EQUATION OF A LINE AND A PLANE

24. Line passing through the points $2\bar{a} + 3\bar{b} - \bar{c}, 3\bar{a} + 4\bar{b} - 2\bar{c}$ intersects the line passing through the points $\bar{a} - 2\bar{b} + 3\bar{c}, \bar{a} - 6\bar{b} + 6\bar{c}$ at P. Position vector of P =

1) $2\bar{a} + \bar{b}$ 2) $\bar{a} + 2\bar{b}$ 3) $3\bar{a} + 4\bar{b}$ 4) $\bar{a} - 2\bar{b}$

25. P is a point on the line through the point A whose position vector is $\bar{a}$ and the line is parallel to the vector $\bar{b}$. If $PA = 6$, then the position vector of P is

1) $\bar{a} + 6\bar{b}$ 2) $\bar{a} \pm \frac{6}{|\bar{b}|}\bar{b}$ 3) $\bar{a} - 6\bar{b}$ 4) $\bar{b} + \frac{6}{|\bar{a}|}\bar{a}$

LEVEL-II (C.W)-K E Y

01) 3	02) 3	03) 2	04) 3	05) 2	06) 1
07) 3	08) 2	09) 4	10) 3	11) 3	12) 1
13) 1	14) 2	15) 3	16) 3	17) 4	18) 2
19) 1	20) 1	21) 3	22) 2	23) 1	24) 2
25) 2					

LEVEL-II (C.W)-HINTS

1. $\sqrt{p^2+9+16}=\sqrt{q+25}\Rightarrow p^2=q$

2. $\overline{OA}=\bar a+\bar b$, $\overline{OC}=\bar a+k\bar b$

$\overline{OB}=\bar a-\bar b$, $\overline{AB}=-2\bar b$

$\overline{AC}=(k+1)\bar b$ $\Rightarrow$ k is any real number

3. $\bar f\left(\frac{5}{4}\right)=\bar i-\frac{1}{4}\bar j+2\bar k$

4. $\overline{AB}=-a-b+(x-3)\bar c$

$\overline{AC}=(y-2)\bar a-4\bar b-3\bar c$

$Now,\frac{y-2}{-1}=\frac{-4}{-1}=\frac{-3}{x-3}\Rightarrow y=-2,\ x=9/4$

5. Eliminate $\bar p$ and compare $|\bar p|$ and $2|\bar q|$.

6. $\begin{vmatrix}0 & 0 & 2\lambda-1\\ 0 & \lambda & 4\\ 1 & 2 & 3\end{vmatrix}=0\Rightarrow(2\lambda-1)\{0-\lambda\}=0$

$\Rightarrow\lambda=0,\ \lambda=\frac{1}{2}$, Vectors are coplanar for values of $\lambda=0,\frac{1}{2}$, Vectors are non coplanar for all except two values of λ.

7. $AB^2=BC^2+CA^2$

$\Rightarrow 1+4+36=(a-1)^2+36+(a-2)^2+4$

$\Rightarrow 41=2a^2-6a+45\Rightarrow 2a^2-6a+4=0$

$\Rightarrow a^2-3a+2=0$

$\Rightarrow(a-1)(a-2)=0\Rightarrow a=1,2$

8. $\overline{AC}=\frac{1}{3}\overline{AO}=-\frac{\bar a}{3}$, $\overline{CD}=3\overline{OB}=3\bar b$

$\overline{AD}=3\bar b-\frac{\bar a}{3}$

9. $\overline{OC}=\frac{\overline{OA}+2\overline{OB}}{3}$

10. Let point A be taken as origin . Then the position vectors of B,C and D are $\bar a,\bar a+\bar b$ and $\bar b-\bar a$ respectively .

P.V. of $M=\frac{\bar a+\bar b+\bar a}{2}=\bar a+\frac{\bar b}{2}$

$\Rightarrow DN:NM=4:1$

$P.V.\ of\ N=\frac{4\left(\bar a+\frac{\bar b}{2}\right)+\bar b-\bar a}{5}=\frac{3}{5}(\bar a+\bar b)$; $\overline{AN}=\frac{3}{5}\overline{AC}$

11. Take 'A' as origin and let $\overline{AB}=\bar b$ and $\overline{AC}=\bar c$

$\therefore D=\frac{\bar b+\bar c}{2},E=\frac{\bar b+\bar c}{4}$, the equation of the line AC is $\bar r=t\bar c$.... (1) and the equation of the line BE is $\bar r=(1-s)\bar b+s\left(\frac{\bar b+\bar c}{4}\right)$........(2)

At the common point F, $1-s+\frac{s}{4}=0$ and $t=\frac{s}{4}$ $\therefore s=\frac{4}{3}$ and $t=\frac{1}{3}$ putting the value of t in (1), $\overline{AF}=\frac{1}{3}\overline{AC}$ $\therefore\overline{AF}:\overline{FC}=1:2$

12. $\bar r=\lambda\bar a+\mu\bar b+v\bar c$

Compare like vectors, $\mu=1,v=2,\lambda=3$.

Hence , $\mu,\frac{\lambda}{2},v$ are in A.P.

13. $\overline{AB}=|\overline{AB}|.\left(D.C.s\ of\ \overline{AB}\right)$

$\Rightarrow 2\sqrt{6}=t\sqrt{6}\Rightarrow t=2$

14. $|\bar e|=10$ and $\alpha=\beta=\gamma$

15. Use transformation of axes

16. $\bar\alpha=\bar a+\bar b+\bar c=6\bar i+12\bar j$

let $\bar\alpha=x\bar a+y\bar b$. Then

$6x+2y=6$ and $-3x-6y=12$

17. $\bar r=\frac{|\overline{BC}|\bar a+|\overline{CA}|\bar b+|\overline{AB}|\bar c}{|\overline{BC}|+|\overline{CA}|+|\overline{AB}|}$

18. Origin is the circumcentre and the orthocentre is $\bar a+\bar b+\bar c=9\bar i+9\bar j+12\bar k$

$\therefore$ The distance between circumcentre and orthocentre is $|\bar a+\bar b+\bar c|$.

19. Follow synopsis

20. Let $G(2,-1,2) \Rightarrow OG = \sqrt{1+4+4}$

$|\overline{OG}'| = 3+1$

21. $D; \overline{BC} = AB : AC$

22. AB = 6, AC = 3 , D divides $\overline{BC}$ the ratio 2:1

23. Let O be the centre.

$\overline{A_1A_{i+1}} = \overline{OA_{i+1}} - \overline{OA_i}$

$\sum_{i=1}^{n-1} \overline{A_1A_{i+1}} = \sum_{i=1}^{n-1}\left[\overline{OA_{i+1}} - \overline{OA_i}\right] = \bar{0}$

24. On verification $\bar{a}+2\bar{b}$ common point to both the lines.

25. Equation of line $\bar{r} = \bar{a} + t\bar{b}$

$PA = 6 \Rightarrow |\bar{r}-\bar{a}| = 6 \Rightarrow |t\bar{b}| = 6 \Rightarrow t = \pm\frac{6}{|\bar{b}|}$

LEVEL - II (H.W)

POSITION VECTOR,UNIT AND EQUAL VECTORS:

1. Let us define, the length of a vector as $|a|+|b|+|c|$. This definition coincides with the usual definition of the length of a vector $a\bar{i}+b\bar{j}+c\bar{k}$ iff

1) $a=b=c=0$ 2) Any one of a,b,c is zero
3) any two of a,b,c are zero 4) $a=b=c\neq 0$

COLLINEAR,LIKE AND UNLIKE VECTORS:

2. P,Q and R are three points with position vectors $\bar{i}+\bar{j}, \bar{i}-\bar{j}$ and $a\bar{i}+b\bar{j}+c\bar{k}$ respectively. If P,Q and R are collinear, then

1) $a=b=c=0$
2) $a=1$, b, c are any real numbers
3) $a=b=c=1$
4) $a=1$, $c=0$ and b is any real number

3. If $10\bar{i}+3\bar{j}$, $12\bar{i}-5\bar{j}$ and $\lambda\bar{i}+11\bar{j}$ are the position vectors of three collinear points. Then λ is

1) 4 2) 8 3) 12 4) 22

4. Let a=(1,1,-1), b=(5,-3,-3) and c=(3,-1,2). If $\bar{r}$ is collinear with $\bar{c}$ and has length $\frac{|\bar{a}+\bar{b}|}{2}$, then $\bar{r}$ equals

1) $\pm 3\bar{c}$ 2) $\pm\frac{3}{2}\bar{c}$ 3) $\pm\bar{c}$ 4) $\pm\frac{2}{3}\bar{c}$

5. The vectors $2\bar{i}+3\bar{j}, 5\bar{i}+6\bar{j}$ and $8\bar{i}+\lambda\bar{j}$ have their initial point at $(1,1)$. The value of λ so that the vectors terminate on one straight line is

1) 3 2) 6 3) 9 4) 12

COPLANAR AND NON COPLANAR VECTORS:

6. If a is a non-zero scalar, then the vectors $\bar{\alpha} = a\bar{i}+2a\bar{j}-3a\bar{k}$, $\bar{p} = (2a+1)\bar{i} + (2a+3)\bar{j}+(a+1)\bar{k}$ and $\bar{r} = (3a+5)\bar{i}+(a+5)\bar{j}+(a+2)\bar{k}$ are

1) coplanar if $a<0$ 2) coplanar if $a>0$
3) always coplanar 4) never coplanar

7. Let $A=2\bar{i}+4\bar{j}-\bar{k}$; $B=4\bar{i}+5\bar{j}+\bar{k}$. If the centroid G of the triangle ABC is $3\bar{i}+5\bar{j}-\bar{k}$ then the position vector of C is

1) $3\bar{i}-6\bar{j}+3\bar{k}$ 2) $3\bar{i}-6\bar{j}-3\bar{k}$
3) $3\bar{i}-6\bar{j}+2\bar{k}$ 4) $3\bar{i}+6\bar{j}-3\bar{k}$

8. $\bar{a}$ and $\bar{b}$ are non collinear vectors. $\bar{u} = x\bar{a}+2y\bar{b}, \bar{v} = -2y\bar{a}+3x\bar{b}$ and $\bar{w} = 4\bar{a}-2\bar{b}$, are vectors such that $2\bar{u}-\bar{v} = \bar{w}$. Then

1) $x=\frac{4}{7}, y=\frac{6}{7}$ 2) $x=\frac{10}{7}, y=\frac{4}{7}$
3) $x=\frac{8}{7}, y=\frac{2}{7}$ 4) $x=6, y=4$

9. A point collinear with (1,-2,-3) and (2,0,0) among the following is

1) (0,4,6) 2) (0,-4,-5) 3) (0,-4,-6) 4) (0,-4,6)

SECTION FORMULA, MID POINT

10. **In the $\Delta OAB, M$ is the mid point of AB, C is a point on OM, such that $2\overline{OC} = \overline{CM}$. X is a point on the side OB such that $\overline{OX} = 2\overline{XB}$. The line XC is produced to meet OA in Y. Then $\frac{OY}{YA} =$**

1) $\frac{1}{3}$ 2) $\frac{2}{7}$ 3) $\frac{3}{2}$ 4) $\frac{2}{5}$

11. **In triangle ABC, D and E are points on BC and AC respectively such that $\overline{BD} = 2\overline{DC}$ and $\overline{AE} = 3\overline{EC}$ Let P be the point of intersection of AD and BE then BP/PE =**

1) $\frac{9}{8}$ 2) $\frac{3}{8}$ 3) $\frac{8}{3}$ 4) $\frac{8}{9}$

LINEAR COMBINATION,LINEARLY INDEPENDENT AND DEPENDENT VECOTORS,DC'S & DR'S

12. **If $\overline{r} = 3\overline{i} + 2\overline{j} - 5\overline{k}$, $\overline{a} = 2\overline{i} - \overline{j} + \overline{k}$, $\overline{b} = \overline{i} + 3\overline{j} - 2\overline{k}$ and $\overline{c} = -2\overline{i} + \overline{j} - 3\overline{k}$ such that $\overline{r} = l\overline{a} + m\overline{b} + n\overline{c}$, then l-m-n =**

1) 0 2) 2 3) 3 4) -2

13. **If $\overline{a} = \overline{i} + \overline{j} + \overline{k}, \overline{b} = 4\overline{i} + 3\overline{j} + 4\overline{k}$, $\overline{c} = \overline{i} + \alpha\overline{j} + \beta\overline{k}$ be linearly dependent vectors and $|\overline{c}| = \sqrt{3}$ then**

1) $\alpha = 1, \beta = -1$ 2) $\alpha = 1, \beta = \pm 1$
3) $\alpha = -1, \beta = \pm 1$ 4) $\alpha = \pm 1, \beta = 1$

14. **x-component of $\overline{a}$ is twice its y-component. If the magnitude of the vector is $5\sqrt{2}$ and it makes an angle of 135^0 with z-axis then the vector is**

1) $(2\sqrt{3}, \sqrt{3}, -3)$ 2) $(2\sqrt{6}, \sqrt{6}, -6)$
3) $(2\sqrt{5}, \sqrt{5}, -5)$ 4) $(\sqrt{6}, \sqrt{6}, -6)$

GEOMETRICAL APPLICATIONS

15. $A_2 = a_2 \cos\alpha + a_3 \sin\alpha = 8,$
$A_3 = -a_2 \sin\alpha + a_3 \cos\alpha = -4$ **with respect to rectangular coordinate system OXYZ, the components of vector $\overline{a} = (2,4,8)$. The system is rotated through an angle of $\frac{\pi}{2}$ about X-axis. The components of $\overline{a}$ in the new system are**

1) (2, 8, 4) 2) (2, 8, -4)
3) (2, -8, 4) 4) (2, -8, -4)

16. **A vector $\overline{a}$ has components 2p and 1 with respect to a rectangular cartesian coordinate system. This system is rotated through a certain angle about the origin in the counter clock - wise sense. If, with respect to the new system, $\overline{a}$ has components p + 1 and 1 then**

1) $p = 0$ 2) $p = 0$ or $p = -1/3$
3) $p = -1$ or $p = 1/2$
4) $p = 1$ or $p = -1/3$

17. **If $\overline{a}, \overline{b}, \overline{c}, \overline{d}$ are the position vectors of four points A, B, C, D in a plane such that $|\overline{a} - \overline{d}| = |\overline{b} - \overline{d}| = |\overline{c} - \overline{d}|$ then the point D is the ... of ΔABC**

1) centroid 2) circumcentre
3) orthocentre 4) incentre

18. **If $\overline{p}$ is the position vector of the orthocentre and $\overline{g}$ is the position vector of the centroid of the triangle ABC when circumcentre is the origin if $\overline{p} = \lambda\overline{g}$ then $\lambda =$**

1) 3 2) 2 3) $\frac{1}{3}$ 4) $\frac{2}{3}$

19. **The position vectors of A,B are $\overline{a}, \overline{b}$ respectively and the position vector of C is $\frac{\overline{a}}{2} + \frac{\overline{b}}{3}$.Then**

1) C is inside the ΔOAB
2) C is outside the ΔOAB but inside the angle AOB
3) C is outside the ΔOAB but inside the angle OAB
4) can not be said

20. In a trapezium ABCD, $\overline{BC} = \lambda\overline{AD}$ and $\overline{x} = \overline{AC} + \overline{BD}$. If $\overline{x} = p\overline{AD}$ then $p =$

1) $\lambda - 1$ 2) $\lambda + 1$
3) $1 - \lambda$ 4) $2\lambda - 1$

ANGULAR BISECTORS

21. Taking 'O' as origin the position vectors of A,B are $\overline{i} + 3\overline{j} - 2\overline{k}, 3\overline{i} + \overline{j} - 2\overline{k}$ The vector $\overline{OC}$ *is* bisecting the angle AOB and 'C' is a point on line $\overline{AB}$ then $\overline{OC}$ is

1) $4(\overline{i} + \overline{j} - \overline{k})$ 2) $2(\overline{i} + \overline{j} - \overline{k})$
3) $(\overline{i} + \overline{j} - \overline{k})$ 4) $6(\overline{i} + \overline{j} - \overline{k})$

22. If the vertices of a ΔABC are A = (1,-1,-3), B = (2,1,-2) and C = (-5,2,-6), then the length of the internal bisector of angle A is

1) $\frac{3\sqrt{10}}{2}$ 2) $\frac{3\sqrt{10}}{5}$ 3) $\frac{3\sqrt{10}}{7}$ 4) $\frac{3\sqrt{10}}{4}$

REGULAR POLYGON

23. If ABCDE is a regular pentagon and $\overline{AB} + \overline{AE} + \overline{BC} + \overline{DC} + \overline{ED} = (\lambda - 1)(\overline{AC})$ then $\lambda =$

1) 1 2) 2 3) 3 4) 4

VECTOR EQUATION OF LINE AND PLANE

24. P, Q, R and S are four points with the position vectors $3\overline{i} - 4\overline{j} + 5\overline{k}$, $4\overline{k}$, $-4\overline{i} + 5\overline{j} + \overline{k}$ and $-3\overline{i} + 4\overline{j} + 3\overline{k}$ respectively. Then the line $\overline{PQ}$ meets the line $\overline{RS}$ at the point [EAM-2013]

1) $3\overline{i} + 4\overline{j} + 3\overline{k}$ 2) $-3\overline{i} + 4\overline{j} + 3\overline{k}$
3) $-\overline{i} + 4\overline{j} + \overline{k}$ 4) $\overline{i} + \overline{j} + \overline{k}$

25. A line passes through the points whose position vectors are $\overline{i} + \overline{j} - 2\overline{k}$ and $\overline{i} - 3\overline{j} + \overline{k}$. The position vector of a point on it at unit distance from the first point is

1) $\frac{1}{5}\left(5\overline{i} + \overline{j} - 7\overline{k}\right)$ 2) $\frac{1}{5}\left(4\overline{i} + 9\overline{j} - 15\overline{k}\right)$
3) $(\overline{i} - 4\overline{j} + 3\overline{k})$ 4) $\frac{1}{5}(\overline{i} - 4\overline{j} + 3\overline{k})$

LEVEL - II (H.W)-KEY

01) 3	02) 4	03) 2	04) 3	05) 3	06) 4
07) 4	08) 2	09) 3	10) 2	11) 3	12) 1
13) 4	14) 3	15) 2	16) 4	17) 2	18) 1
19) 1	20) 2	21) 2	22) 4	23) 3	24) 2
25) 1					

LEVEL - II (H.W)-HINTS:

1. $\sqrt{a^2 + b^2 + c^2} = |a| + |b| + |c|$ is possible only when any two of them are zero

2. $\overline{AB} = -2\overline{j}$, $\overline{AC} = (a-1)\overline{i} + (b-1)\overline{j} + c\overline{k}$ and $\text{AB} = t\,\overline{AC}$

3. $\begin{vmatrix} \lambda & 11 & 1 \\ 10 & 3 & 1 \\ 12 & -5 & 1 \end{vmatrix} = 0$

4. We have, $\overline{r} = \lambda\overline{c}$, Given $|\overline{a} + \overline{b}| = 2|\lambda||\overline{c}|$

5. $\begin{vmatrix} 2 & 3 & 1 \\ 5 & 6 & 1 \\ 8 & \lambda & 1 \end{vmatrix} = 0$

6.

7. C=3G-A-B

8.

9. let P(x,y,z) be collinear with the given points. Then

10.

equation of is

..... (1)

equation of the line is, ... (2)

From (1) and (2) and ,

or ,

11. Let and be the position vectors of A,B,C,D and E points respectively.
.......(i), ...(ii)
From (i) & (ii) , we get
of P.
Then
12.
find
13.
14. Let
Also
Required vector
15. Use transformation of axes
16.
17.
is equidistant from the points A, B, C
18. Centroid G divides line joining orthocentre and circumcentre in the ratio 2 : 1
19.
Then C lies internally in OP where P is a point which divides AB in the ratio 3 : 5 externally
20.
21.
22.
23.
24. Eq. of the line is
Eq. of is
25.

LEVEL - III

1. A unit tangent vector at t = 2 on the curve is

1) 2)

3) 4)

2. The vectors , are collinear for :

1) Unique value of

2) Unique value of

3) No value of x

4) Infinitely many values of

3. Let a line cut the sides PQ,PS and diagonal PR of a parallelogram at and respectively such that and , then

1) 2)

3) 4)

4. The value of so that the points P, Q, R, S on the sides OA, OB, OC and AB of a regular tetrahedron are coplanar. When and is

1) 2) 3) 4)

5. If and where k > 0 and X, Y are the midpoints of DB and AC respectively such that and , then k =

1) 2) 3) 4)

6. 'O' is the origin in the Cartesian plane. From the origin 'O' take point A in the North-East direction such that =5, B is a point in the North-West direction such that .

Then is

1) 25 2) 3) 4)

7. Let O be the origin of the coordinate system in the Cartesian plane, and be vectors making angles and respectively with the positive directions of the X-axis (i.e., in the counter clock wise). Rectangle OPQR is completed and M is the midpoint of PQ. If the line meets the diagonal PR at T, and then is

1) 2)

3) 4)

8. If is the vector whose initial point divides the joining and in the ratio : 1 and terminal point is at origin. If then

1)

2)

3) 4)

9. Let AC be an arc of a circle, subtending a right angle at the centre O. The point B divides the arc AC in the ratio 1:2 If and , then in terms of and is

1) 2) 3) 4)

10. In , E is the mid point of AB and F is a point on OA such that . If C is the point of intersection of OE and BF, then find the ratios OC : CE and BC : CF are

1) 2)

3) 4)

11. ABCD is a parallelogram and P is a point on the segment dividing it internally in the ratio 3:1 the line meets the diagonal in Q. Then the ratio AQ:QC is

1) 3:4 2) 4:3 3) 3:2 4) 2:3

12. In , are points on respectively, dividing them in the ratio 1:4, 3:2 and 3 : 7. The point S divides AB in the ratio 1:3.

Then

1) 2) 3) 4)

13. In a triangle ABC, if then the vector of magnitude units directed along , where O is the circumcentre of triangle ABC is

1) 2)

3) 4)

14. Vectors are laid off from one point.Vector ,which is being laid off from the same point dividing theagle between vectors and in equal halves and having the magnitude ,is

1) 2)

3) 4)

15. **3 forces are applied to a vertex of a cube which are 1, 2 and 3 in magnitude and are directed along the diagonals of the faces of the cube meeting in that vertex. The magnitude of the resultant of these forces is**

1) 3 2) 4 3) 5 4) 6

16. **A man starts at the origin of the coordinate axes and walks a distance of 3 units in the North-East direction and then walks distance of 4 units in the North-West direction to reach the point P. Then equals**

1) 2)

3) 4)

17. **ABCDEF be a regular hexagon in the xy plane and Then**

1) 2)

3) 4)

18. **OABCDE is a regular hexagon of side 2 units in the xy-plane O being the origin and OA taken along the x-axis. A point P is taken on a line parallel to z-axis through the centre of hexagon at a distance of 3 units from O. The vector is**

1) 2)

3) 4)

19. **The triangle ABC is defined by the vertices A = (0,7,10), B = (-1,6,6) and C = (-4,9,6) let D be the foot of the altitude from B to the side AC . Then is**

1) 2)

3) 4)

20. **The equation to the altitude of the triangle formed by (1, 1, 1), (1, 2, 3), (2, -1, 1) through (1, 1, 1) is**

1)

2)

3)

4)

21. **Image of the point P with position vector in the line whose vector equation is has the position vector**

1) 2)

3) 4)

22. **If and then the vector equation of the plane parallel to and passing through the centroid of the tetrahedron ABCD is**

1)

2)

3)

4)

LEVEL - III-KEY

01) 2	02) 2	03) 2	04) 4	05) 2	06) 2
07) 4	08) 1	09) 3	10) 2	11) 1	12) 2
13) 1	14) 1	15) 3	16) 4	17) 2	18) 1
19) 2	20) 3	21) 2	22) 3		

LEVEL - III-HINTS

1.

2. For collinearity,

now, draw the graphs of and ,
then observe

3. (parallelogram law)

Since and points are collinear,

4.

,

, ,

P, Q, R, S are coplanar points

are coplanar vector

5.

6.

Then

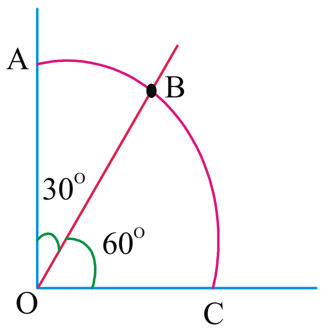
7.

Now

Now

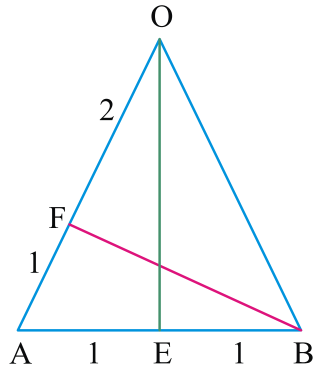
8.

9.



Given that and are coplanar , then can be written as the linear combination of two non-colinear vectors and . Let and be taken a long x and y axes respectively , and where r be the radius, then we can write

10.



11. Take and so that and . Equation of the line is and the equation of the line is

Equating the coefficients of

and hence

12.

13. Right angled triangle.

Vector of magnitude units along

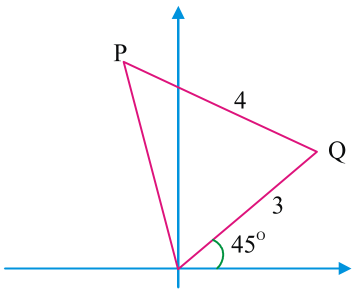
14. bisecting the angle between is

Required vector =

15. Let the diagonals of the faces of the cube are

Now

16. Let



17. Take A as origin then

18. Let be the centre of the hexagon
Let M be the mid point of OA
OM = 1, and

and

19. B divides in the ratio BA : BC

20. Let A ,

Then

Midpoint of BC is and

,

Equation of is

21. Let (x, y, z) is image of (7, –1, 2) w.r.t. the given line.Then ... (1)

and .... (2)

Solving (1) or (2)

22. Centroid ,

Equation of plane is

LEVEL - IV

Assertion-Reason Type :

Note :

a) Both A and R are true and R is a correct explanation of A
b) Both A and R are true and R is not a correct explanation of A
c) A is true but R is false.
d) A is false but R is true

1. **Assertion (A): If are collinear and is a non zero vector then are coplanar.**
Reason(R): Two vectors are always coplanar.
1) a 2) b 3) c 4) d

2. **Observe the following statements**
Assertion (A): Three vectors are coplanar if one of them is expressible as a linear combination of other two
Reason(R): Any three coplanar vectors are linearly dependent then which of the following is true
1) a 2) b 3) c 4) d

3. **Assertion (A): If ,**
,
are linearly dependent and
= then
Reason(R): For coplanar vectors every vector can be expressed as linear combination of other.
1) a 2) b 3) c 4) d

4. **Assertion (A): If vectors and are non collinear then the lines**
,
are coplanar.
Reason(R): There exist and such that the two values of become same.
1) a 2) b 3) c 4) d

5. **Assertion (A): If are non coplanar then , , are coplanar.**
Reason (R): The system of equations , , has non zero solutions.
1) a 2) b 3) c 4) d

6. **Statement-1: If be a line, then the point on the line such that its magnitude 15 units from the point (3, 1, -1) is .**
Statement-2: The vector equation of the plane passing through (0, 0, 0), (0, 5, 0) and (2, 0, 1) is ,
1) only I 2) only II
3) Both I and II 4) Neither I nor II

7. **Statement-1: If OABC is a parallelogram and then the equation of the side is ,**
Statement-2: Vector equation of the median through A of triangle ABC with are respectively position vector of the vertices A, B, C is .
Which of the above statements is/are true.
1) only I 2) only II
3) Both I and II 4) Neither I nor II

8. **If D,E,F are the midpoints of sides BC, CA, AB of triangle ABC and G is the centroid then match the following**
1) a)
2) b)
3) c) O
4) d)
e)
1) c,d,a,b 2) d,b,a,c 3) c,d,e,a 4) b,c,e,a

9. **In are midpoints of the sides BC, CA and AB respectively. 'O' is the circumcentre 'G' is the centroid. 'H' is the orthocentre and P is any point.**

Match the following
1) a)
2) b)
3) c)
4) d)
1) a, b, c, d 2) c, a, b, d
3) c, a, d, b 4) a, b, d, c

LEVEL - IV-KEY

01) 1	02) 2	03) 1	04) 1	05) 2
06) 3	07) 3	08) 1	09) 3	

LEVEL - IV-HINTS

1. If are coplanar then also lie on the same plane.
2. Both A and R are true,
3.
4. compare the coefficients of and . Find and .
5. In both cases
6. A = (3, 1, -1) and , and the point P = for II, the equation of the plane passing through origin, is
7. For I, the line parallel to and passing through . For II, The medium passing through &
8. ,
9. Let ,